

Capital Markets Questions

Debt & Leveraged Finance

What is the difference between a bond and a leveraged loan?

While some features of leveraged loans and bonds can overlap, the key difference is that a loan is a private transaction between a borrower and a lender. The lender is a single bank or a small syndicate of banks or institutional investors. The interest cost is often LIBOR plus a spread, and the loan is often secured by collateral with strict covenants, while the repayment of principal can happen over time or as a bullet payment at the end. Given the collateral, earlier principal repayments, and covenants, loans are less risky and carry lower interest rates than bonds.

Corporate bonds, in contrast, must be registered with the SEC and are public transactions. Bonds are issued to institutional investors and traded freely on the secondary bond market, leading to a broader investor base. Bonds are usually priced at a fixed rate with semi-annual payments, have longer terms than loans, and have a balloon payment at maturity. Since bonds come with less restrictive covenants and are usually unsecured, they're riskier for investors and therefore command higher interest rates than loans.

Putting it all together, below is a table outlining the typical features of debt used in leveraged finance:

	Leveraged loans		Bonds		
Debt Type	Revolver	Term Loan A (Bank Debt); Term Loan B/C/D (Institutional)	Senior secured	Senior unsecured	Subordinated
Lender	Institutional investors & banks		Institutional investors		
Coupon	Floating, i.e. LIBOR + 300 bps		Fixed, i.e. 8.00% coupon paid semi-annual		
Cash/PIK interest	Cash interest				Cash or PIK
Interest rate	Lowest <-----> Highest				
Principal repayment schedule	None	Some principal amortization	Bullet at end of term		
Secured/unsecured	Secured (1st and 2nd liens)			Unsecured	
Priority in bankruptcy	Highest<----->Lowest				
Term	3-5 years	5-7 years	5-10 years		
Covenants	Mostly incurrence ("covenant lite"); Some maintenance (strictest)		Incurrence		
Call protection	No			Yes	

Read More → [Ultimate Guide to Debt & Leveraged Finance](#)

What is the difference between investment-grade and speculative-grade debt?

- **Investment-Grade Debt:** Investment-grade debt (often called high-grade debt) has a credit rating above BBB/Baa. This category of debt is issued by companies with a strong credit profile. Investment-grade debt is considered safe, given the low risk of default.
- **Speculative-Grade Debt:** Speculative-grade debt has a credit rating below BB/Ba. These types of debt are issued by more leveraged companies with a riskier credit profile. Given this increased risk of default and bankruptcy, the interest rates on these riskier debts will be significantly higher to compensate investors for taking on the additional risk.

What does it mean when a debt tranche is denoted as 1st lien or 2nd lien?

Lien is defined as the seniority and the priority of payment to a debt holder relative to the other tranches. A lien is a legal claim against the assets of a borrowing company (i.e., used as collateral) and the right to seize those assets first in forced liquidation/bankruptcy scenarios.

- **1st Lien Debt:** The highest seniority, 1st lien, is fully secured by the company's assets and has the first claim to collateral in a liquidation/bankruptcy scenario (e.g., revolver, term loans).
- **2nd Lien Debt:** Right below 1st lien loans sits 2nd lien where compensation is provided only if there's collateral value remaining once 1st lien lenders are repaid in full. These debt types are riskier and more expensive for borrowers (e.g., high-yield bonds, mezzanine financing).

Tell me about the different classifications of term loans.

- **Term Loan "A":** TLAs refer to secured loans syndicated to banks and are typically packaged alongside a revolving credit facility. TLAs have shorter terms (~5 years), carry higher amortization levels than other term loans, and are amortized evenly over their tenor (i.e., "straight-lined").
- **Term Loan "B," "C," or "D":** An institutional term loan (B/C/D) is a loan facility syndicated to institutional, non-bank investors such as hedge funds, CLOs, mutual funds, and insurance companies. These term loans differ from TLAs in having longer terms while requiring no principal amortization before maturity – instead, they involve nominal amortization with a bullet payment due at maturity. TL B/C/Ds are more prevalent in LBOs than TLAs, as B/C/Ds have less strict covenants, longer terms, and require less principal amortization each year. The "B," "C," or "D" designation is more indicative of the investor base than the priority rank.



Watch Video →
Term Loans
and Revolvers

What is the difference between a secured and unsecured loan?

- **Secured Loans:** If a debt instrument is secured, that means the debt is backed by collateral. The assets of the borrower were pledged as collateral to get favorable financing terms. If the company were to go bankrupt, the lenders have a legal claim on the pledged collateral. Leveraged loans are secured by collateral and are the safest security class for a lender. Most term loans and revolvers in the leveraged loan market are syndicated to institutional investors such as hedge funds, CLOs, and mutual funds.
- **Unsecured Loans:** For unsecured loans, pension funds, mutual funds, insurance companies, hedge funds, and some banks are typically willing to invest in this relatively riskier type of debt for the higher yield.

How are leveraged loans usually priced?

Leveraged loans are usually priced off LIBOR plus a spread. In addition, loans often include a LIBOR floor, so an example would be a pricing of "LIBOR + 3%" (300 basis points) with a LIBOR floor of 2%, so the interest rate can never dip below 5%.

What does LIBOR stand for?

LIBOR stands for “London Interbank Offered Rate,” representing the global standard benchmark used to set lending rates. LIBOR is the rate at which banks lend amongst each other. For lenders of debt instruments with floating rates, the debt pricing will be based on LIBOR, the standard interest rate. However, LIBOR is expected to fade away in use as UK regulators have voiced a desire for LIBOR to be phased out by the end of 2021.

What is SOFR, the expected replacement of LIBOR?

Coming up on the horizon and expected to replace LIBOR eventually, the Secured Overnight Funding Rate (SOFR) is a measure of the borrowing costs of cash collateralized by Treasury securities. Said another way, the SOFR is a Repo-based funding rate of the observed transactions overnight.

In terms of debt terminology, what does the coupon rate mean?

The coupon rate simply refers to the annual interest rate (“pricing”) paid on a debt obligation. The interest expense is based on the outstanding principal amount and is modeled as a percentage of the beginning and ending balance of the relevant debt tranche. In terms of payment dates, senior bank debt pays interest each quarter, whereas most bonds pay interest on a semi-annual basis.

How does the coupon on a bond differ from the yield?

The coupon represents the annual interest rate paid based on the notional principal of the bond, while the yield is the annual return on the bond, including the coupon payment adjusted for the premium or discount of the purchase price when held to maturity. One difference is coupons are fixed for the bond’s term, whereas yields move with the markets.

What does it mean when a bond is trading at a discount, par, or premium?

- **Discount:** Price < 100, Yield is Greater than Coupon
- **Par:** Price = 100, Yield is Equal to Coupon
- **Premium:** Price > 100, Yield Less Than Coupon

What is the difference between a fixed and floating interest rate?

- **Fixed Interest Rate:** A fixed interest rate means the interest expense to be paid is the same regardless of changes to the lending environment. A fixed interest rate is more common for riskier types of debt, such as high-yield bonds and mezzanine financing.
- **Floating Interest Rate:** A floating interest rate is tied to LIBOR plus a specified spread (i.e., LIBOR + 2-4%). This pricing type is seen more often for senior debt tranches (e.g., term loans, revolvers).

When would an investor prefer fixed rates over floating rates (and vice versa)?

If interest rates are expected to fall in the near-term future, investors would prefer fixed rates. However, if interest rates are expected to increase, investors would prefer floating rates.

What are some different debt amortization schedules?

The debt amortization schedule refers to the amount of principal the borrower must repay annually. Compliance with this payment schedule is mandatory and not optional for the borrower.

Types of Debt Amortization Schedules:

- **Bullet Maturity:** The entire loan payment is due at the end of the loan’s lifespan.
- **Straight-Line Amortization:** Principal payments must be repaid in equal installments until maturity.
- **Minimum Amortization:** Entails lesser amounts of annual payments (e.g., ~5-10% per year) – therefore, the entire principal will not have been paid off at maturity.

What is a callable bond and how does it benefit the issuer or borrower?

A callable bond can be redeemed by the issuer prior to its maturity, with the decision being at the issuer's discretion. A callable bond enables the issuing company to pay off the debt earlier if they have more free cash flow remaining in the period and can refinance at lower interest rates.

From the investor's perspective, a callable bond gives more optionality to the issuer, so the debt holders are compensated with higher interest rates (compared to non-callable bonds).

When would the prepayment optionality of certain debt tranches be unattractive to lenders?

Some debt instruments include provisions that enable the borrower to repay some principal ahead of the payment schedule without the incurrence of any financial penalties. However, other lenders may include a call protection feature that prohibits borrowers from prepayment until a pre-specified duration has passed. The reason being that certain lenders prefer to disallow prepayment as it implies the receipt of more interest payments in the future.

For instance, if the borrower pays more principal off early, the annual interest payments (inflows to the lender) in the future are reduced since interest is based on the beginning and ending balance of the debt outstanding.

A bond has a call protection clause of NC/2. What does this mean?

Many HYBs will have call protection clauses that last two or three years (denoted as NC/2 and NC/3, respectively). Some are often NC/L, which means the bond is not callable for the term's entire duration. Once a bond becomes callable, the borrower may repay some (or all) of the debt balance and pay less interest. The caveat is that the prepayment penalties could offset those savings on interest – thus, HYB's classification as an expensive financing source.

Therefore, NC/2 means the bond has call protection for two years. Once this two-year period has passed, the borrower can repay the debt along with the prepayment penalty fee.

What is a revolving credit facility and what purpose does it serve to the borrower?

The revolver refers to a company's revolving line of credit drawn down when the free cash flow being generated is insufficient. The revolver acts as a "corporate credit card" for urgent situations. The borrower typically draws from the revolver to meet its short-term working capital requirements after an unexpected, temporary shortage in liquidity. Ideally, the lender doesn't want the revolver fully drawn frequently as it signals a deterioration in cash flows.

The revolver provides the borrower with the optionality to drawdown, repay, and reborrow on an "as-needed" basis.

What is the undrawn commitment fee associated with revolvers?

A revolver typically comes with a small $< 1\%$ fee, which is an annual fee paid out to the lender. The borrower is charged an annual fee on the unused amounts, called the undrawn commitment fee.

What is the difference between an asset-based loan and a cash flow revolver?

The maximum amount that can be drawn from an ABL revolver is based on the company's liquid assets. Thus, the amount is tied to borrowing-base lending formulas to limit borrowing to a certain percentage of the collateral – most often inventory and accounts receivable (e.g., 80% of A/R + 65% of Inventory)

The maximum amount that can be borrowed for cash flow revolvers is tied to the borrower's historical and projected cash flow generation. Therefore, covenants are more restrictive due to the uncertainty around future cash flows. Unlike physical assets such as inventory, a company's future cash flows cannot be pledged as collateral or seized in bankruptcy, hence its less favorable terms.

Why do revolvers normally not have a leverage test?

In most cases, revolvers will only have an interest coverage ratio test (e.g., $> 2.0 \times \text{EBITDA} / \text{Cash Interest}$) and have the simplest covenant structure relative to other tranches of debt. This is because the revolver has the highest priority in the capital structure and has a priority claim to the borrower's assets.

Therefore, the lender that provided the revolving credit line is unconcerned if the borrower raises additional debt, since this means the company has more cash available (on which the revolver has the first claim). In the scenario that the borrower undergoes bankruptcy, the revolver will almost certainly be made whole.

What is unitranche debt and its benefits?

Unitranche debt financing is a blended hybrid of senior debt and subordinated debt. The distinct characteristics of unitranche debt are that it's just one tranche of debt instead of two (as the name suggests), and the debt is priced at a blended interest rate. Traditionally, there would be 1st and 2nd lien debt, and the borrower would have to get financing from two (or more) different lenders, which could make securing financing packages a time-consuming process. But in recent years, non-bank institutional lenders began to provide the entire package and customize it based on negotiations.

The main advantage to borrowers over traditional credit facilities is that it enables borrowers to have "one-stop-shop" financing (i.e., the convenience factor). The borrowers would have only one set of loan documents, one set of covenants, and a much simpler and faster process to close.

What is the difference between a bond's coupon rate and the bond's current yield?

The coupon rate ("nominal yield") represents a bond's annual coupon divided by its face (par) value. The current yield on a bond equals the bond's coupon payment divided by the bond's price.

For example, a bond trading at 90 with a \$100 face value and a \$6 coupon has a 6% coupon rate and a current yield of 6.7% ($\$6/90$). While the coupon rate is always the same, the current yield fluctuates based on the market price of a bond.

What is the difference between current yield and yield to maturity?

The current yield on a bond equals the bond's coupon payment divided by the bond's price. The current yield is a way to discuss coupon rates when the bond price deviates from par. For example, a bond trading at 90 with a \$100 face value and a \$6 coupon has a 6% coupon rate and a current yield of 6.7% ($\$6/90$).

Unlike YTM, the current yield is not the true yield of a bond as it doesn't capture any yield associated with principal recovery, nor does it assume the reinvestment of coupon payments. The YTM is the internal rate of return of a bond. YTM considers coupon payments, principal recovery, assumes reinvestment at the same rate (an iterative process), and time to maturity.

Could you define fixed income and name a few examples?

Fixed-income securities provide their investors with a stream of fixed periodic interest payments and then the return of principal at the end of its term. The fixed amount of interest is paid in the form of coupon payments, usually semi-annually. While all bonds technically fall under fixed income, fixed income usually refers to low-return, low-risk bonds (as opposed to mezzanine financing and HYBs). Examples include treasury notes, treasury bonds, treasury bills, municipal bonds, money markets, and certificates of deposits (CDs).

What does the money market refer to and what is the typical maturity range?

The money market refers to the purchase and sale of large quantities of short-term bonds and other debt instruments overnight. The maximum maturity of these short-term bonds is 397 days (~13 months), and the investors are usually risk-averse with limited risk appetite.

What are the two main classifications of money market accounts?

1. **Government Money Market Funds:** T-Bills, Discos (Agency Discount Notes, Reverse Repo)
2. **Prime Money Market Funds:** Short-Term Bonds, Commercial Paper, Certificates of Deposits

What is a municipal bond and what is the one distinct benefit it has for investors?

A municipal bond is a debt instrument issued by a state, municipality, or county to finance its capital expenditures needs, including construction needs, road developments, parks, highways, and other public projects. “Munis” could be viewed as loans that investors make to local governments – and for doing so, these bonds are exempt from federal taxes and most state/local taxes.

What is the difference between coupon bonds and discount bonds?

Unlike coupon bonds, zero-coupon bonds (discount bonds) make no payments between issuance and maturity and are priced at a discount to their face value.

What is the difference between Macaulay duration and modified duration?

Macaulay duration is the weighted average timing of the present value of all the cash flows, typically denoted in years. The modified duration, on the other hand, indicates the percentage change in a fixed income instrument's price given a 1% interest rate change – by making a slight adjustment to Macaulay's duration to reflect the price movement given a change in yield.

Asset managers and fixed income investors typically focus on modified duration, as it shows price sensitivity to interest rates. For example, a 10-year bond with a modified duration of 8 would lose 8% in price (say from par or \$100 to \$92) if the yield increased from 1% to 2%.

Read More → [Understanding Duration](#)

What is the relationship between duration and the coupon?

The duration of any coupon-bearing bond will be less than its years to maturity, as there are coupon payments between now and maturity.

- **Lower Coupon → Longer Duration:** The lower the coupon, the longer the duration is and closer the duration is to maturity. The impact of each coupon payment to shorten the payment is reduced.
- **Higher Coupon → Shorter Duration:** The higher the coupon, the shorter the duration. Each coupon payment has a higher cash flow, which shortens the duration.

What is convexity used to measure?

Convexity is a measure of the relationship between the price of a bond and its yield. Often, convexity is used to track the overall exposure risk on a portfolio of bond holdings. In general, a bond with higher convexity would have a higher price, and its addition to a portfolio should increase the systematic risk of the entire portfolio of bonds. Therefore, a high convexity portfolio would be more sensitive to interest rate fluctuations.

What are the three types of covenants found in lending agreements?

- **Affirmative Covenants:** These covenants require that the obligor (borrower) of the debt to perform certain specific tasks. Examples of affirmative covenants are meeting financial reporting requirements, paying taxes, being insured, maintaining licenses/permits, and legal compliance.
- **Negative Covenants:** These covenants restrict the obligor of the debt to refrain from certain specific tasks such as issuing dividends, raising debt, acquiring another company, or pledging assets as collateral.
- **Financial Covenants:** For financial covenants, there are two types: 1) maintenance and 2) incurrence covenants. Maintenance covenants are usually included in credit agreements (bank loans), while incurrence covenants are included in indentures (bonds).

What is the purpose of covenants in debt financing?

Covenants are contractual agreements between lenders and borrowers meant to protect the interests of the lending parties. Failure to comply with these covenants or breaching a covenant can cause the borrower to be placed into default, allowing lenders to seize the borrower's assets. Debt-holders desire assurance of the full receipt of their due payments with a high level of certainty and restricting the borrower into making only risk-averse decisions and maintaining healthy credit statistics decreases the risk of not being paid back.



**Watch Video →
Debt Covenants**

What are maintenance covenants and provide some examples?

The objective of maintenance covenants is to ensure the borrower maintains sufficient profitability and cash flows to service debt payments. Compliance with maintenance covenants is tested each quarter.

Examples of Maintenance Covenants

- Total Debt/EBITDA must remain below 5.0x
- Debt/Equity must never exceed 2.5x
- Interest Coverage Ratio cannot fall below 3.0x

These parameters will change depending on the prevailing market conditions and be industry specific.

What are incurrence covenants and give some examples?

The purpose of incurrence covenants is to prevent the borrower from taking specific actions that could put the lender's payback at risk. Compliance with incurrence covenants is tested when taking a specific action (e.g., new debt issuance, dividends, acquisition).

Examples of Incurrence Covenants

- Restricted from making acquisitions or divesting one of its business segments (or major assets)
- Prevented from raising additional debt, especially if it has higher seniority than the covenant holder
- Cannot distribute dividends to equity shareholders without the approval of the lenders

What is one key difference between maintenance and incurrence covenants?

Maintenance covenants differ from incurrence covenants in that they're subject to periodic tests, typically completed at the end of each quarter. The borrower must routinely prove its compliance with its maintenance covenants to avoid default. Incurrence covenants are tested only once a certain action or event "triggers" it.

Leveraged loans have become increasingly "covenant-lite," what does this entail?

Secured 1st lien loans have shifted towards becoming "covenant-lite" in recent times, meaning they're packaged with less restrictive incurrence covenants than traditional maintenance covenants. Compliance tests of covenant-lite loans are thus only required if the borrower takes a certain action (e.g., raises more debt, dividend issuance), rather than being tested periodically as they traditionally were.

This trend is due to large banks needing to make their financing packages more attractive to borrowers to compete with newer types of institutional lenders that offer more flexibility (e.g., private credit funds, increased usage of unitranche debt).

What are subordinated notes?

Subordinated notes are characterized by longer tenors and higher-interest rates that compensate investors for undertaking more risk. This layer of debt enables the borrower to increase leverage beyond what risk-averse institutional banks will provide. While subordinate notes have a higher cost of capital relative to bank debt and don't allow prepayments, these notes come with less restrictive covenants.

What are the characteristics of mezzanine financing?

Mezzanine financing refers to the layer of financing that lies in between traditional debt and common equity. This category is the lowest form of debt in the capital structure and includes preferred stock, convertible debt, bonds coupled with warrants. All mezzanine debt is unsecured and will be of smaller magnitude relative to the other parts of the capital structure.

The debt terms involved in mezzanine financing are highly negotiated, flexible, and tailored to meet the specific needs of both parties. The interest rates are the highest compared to other less risky tranches and debt, with the option for interest to be paid-in-cash or payment-in-kind (PIK). Also, the conversion feature that some of these securities carry provides the holder with the optionality to partake in the upside potential of the equity (and create dilution for common shareholders).

What are bridge loans?

Bridge loans provide interim financing should the required debt commitments not be available by the closing of the deal (i.e., the borrower could not secure a firm commitment letter from lenders). Investment banks that can do so will provide this type of bridge loan commitment to prevent a transaction from stalling and give assurance that sufficient funding will be available to close the deal.

What is staple financing and which type of lender provides it?

Staple financing is when the sell-side investment bank can provide some initial debt, but rather than holding onto the debt, they usually sell it to debt-oriented hedge funds or institutional investors.

Staple financing is an option available when the sell-side advisor is a large, more institutionalized investment bank that has a balance sheet such as J.P. Morgan, Goldman Sachs, and Morgan Stanley. For clarification, the phrase “has a balance sheet” means the investment bank is diversified in terms of its sources of revenue and has a lending division, as opposed to purely offering M&A advisory services. This is an advantage that bulge bracket banks have over elite boutiques such as Evercore and Qatalyst Partners, where most of their revenue comes from M&A advisory – but this creates the potential for a conflict of interest since the firm serves both sides of the transaction (the seller and the buyer).

What is a firm commitment letter?

Once a financial sponsor or a strategic buyer has met with the target's management team and has proceeded with submitting an LOI, the next step is to get a firm commitment letter from its lender(s). This is done to gain credibility as a buyer, as the most credible bids are the ones with financing commitments prepared with the initial debt terms outlined.

How does a highly confident letter differ from a firm commitment lender?

From both the buyer and seller perspective, committed financing is preferred as the lending bank is practically guaranteed to fund it upon closing. Alternatively, highly confident letters mean the lender believes it can raise the amount of capital required, but they'll not commit to it (and not backstop it with their balance sheet).

Other than covenants, what provisions can be included to protect lender interests?

Besides the traditional covenants found in lender agreements, there'll often be conditions listed that enable the lender to renege on its initial commitment without facing monetary consequences. These provisions are included in the case an unexpected event occurs. The lender doesn't want to be committed to lending in a situation when everything appears to be headed downhill and the risk of default has significantly increased.

In these commitment letters, you would usually see a provision within the agreement to lend as long as certain conditions are met. In nearly all cases, the letter includes phrasing along the lines of how there cannot be a “significantly material adverse effect” that places repayment into question.

What is commercial paper and which types of companies issue them?

Commercial paper is issued by large corporations with high credit ratings as short-term borrowing to finance their working capital needs. The typical commercial paper term is ~270 days, and the debt is issued at a discount (i.e., zero-coupon bond) as an unsecured promissory note. The minor concern associated with commercial paper is that they're unsecured, meaning they're not backed by any collateral. Therefore, only large corporations issue this type of debt as they have strong credit ratings.

Besides market risk, name some other risks that bond investors face?

- **Default Risk:** Default risk means the borrower's creditworthiness has decreased from deteriorating financial performance, and most often, it coincides with a downgrade from a credit agency. The higher the credit rating, the lower the probability of default.
- **Liquidity Risk:** Next, liquidity risk refers to when the ability of the bond to be exchanged has unexpectedly decreased. This means that the number of potential buyers in the market has diminished (e.g., changing lending conditions, unexpected company-specific circumstances).
- **Event Risk:** Finally, event risk pertains to specific events such as M&A or legal investigations that could change the risk of holding a particular bond. For example, if the SEC announces they have started an investigation into a company for fraud, this would have an immediate negative effect on all stakeholders.

Tell me about the debt lender and equity investor "story disconnect."

This disconnect between debt lenders and equity investors is due to lenders being "backward-looking" and focused on how the company has performed historically.

Particular attention is paid to signs of cyclical performance in the borrower's performance and how it was affected by the latest recession. This is because debt lenders are more concerned with the downside case. After all, even if the company performs well, the lenders don't get to participate in the upside alongside the equity investors.

Thus, the continued generation of the borrower's cash flows to satisfy debt obligations is of the utmost highest importance to debt investors.

In the best case scenario, lenders receive all interest payments and repayment of debt principal, whereas equity owners theoretically have unlimited upside.

When a company is raising capital, why is the lender's case usually less optimistic than the projections shown to raise interest from equity investors?

A distinction when dealing with lenders is that there's little incentive to show aggressive assumptions. Thus, lender case forecasts are lower than the base case for the debt covenants to be set off a lower base.

While they must show that the company can meet the debt obligations to be approved for financing, there's no rational reason for a sponsor to stretch its assumptions beyond this.

Debt covenants are usually based on this metric of EBITDA – thus, the EBITDA provided to lenders is the starting point. A higher EBITDA as the starting point means a higher EBITDA level is required to abide by the covenants. Then, future covenants will be based on quarterly projections of this initial EBITDA.

Therefore, the borrower should desire a sufficient "cushion" to operate where, even if its EBITDA were to drop in an economic downturn, the covenant would still not be breached.

What does refinancing risk involve?

Refinancing risk is the concern that a borrower cannot refinance its debt obligations at the same (or similar) rate as before. For example, the credit rating of the company may have been demoted, and lenders are no longer willing to refinance at the same rates.

From the perspective of an investor, what is the interest rate risk in reference to?

Interest rate risk becomes a consideration for bonds with longer maturities. For instance, the market rates constantly change, and the lending environment could turn more favorable (interest rates could decrease).

If an investor is locked into a long-term bond, they'll face more interest rate risk due to the longer maturities. Therefore, the investor would demand a higher yield to compensate for the additional risk taken on by agreeing to a long-term arrangement.

What is prepayment risk?

Prepayment risk (often called reinvestment risk) is the risk that a lender takes on when allowing the borrower to pre-maturely pay down a particular debt. For example, if a borrower has pre-paid their loan and the lender has received the initial principal back – the lender must search for a new borrower.

However, there's the risk that the credit market has become less favorable to lenders, and the lender may not achieve the same yield as before. For this reason, many lenders will not allow the optionality for prepayment, or if they do, they'll attach prepayment penalties as compensation for taking on the risk of having to reinvest at the current market rates (which could be significantly lower).

In mezzanine financing, what is an "equity kicker"?

An equity kicker is often issued as a "deal sweetener" for new debt issuances. Through the inclusion of an equity kicker, mezzanine investors can often increase returns by an extra 100-200 basis points. In return, the cost of financing can be brought down to allow the borrowing company to secure better terms.

"Equity Kicker" Types

- **Warrants:** Warrants function similar to employee stock options such that the mezzanine investors have the option to exercise their options and turn them into common stock if profitable. This usually amounts to 1-2% of the total equity of the borrower.
- **Co-invest:** Mezzanine investors may seek the right to co-invest equity alongside the controlling shareholder, such as the financial sponsor in the case of funding an LBO.
- **Conversion Feature:** If the debt or preferred stock is structured as convertible, the investor has the option to 1) participate in the common equity's upside or 2) continue to receive interest or dividends.

Equity Financing

If a startup is “bootstrapping,” what does this mean?

If a startup is currently in the “bootstrapping” stage, operations are currently being funded entirely by the cash flows generated by the business and through the out-of-the-pocket funds from the entrepreneur. This term is often closely aligned with the concept of a startup remaining “lean” and holding off on raising outside, “unnecessary” capital to first maximize the utilization of their resources (e.g., human capital, intellectual property, product, and market strategy ideas/plans).

The startup is likely attempting to minimize expenses while building its cash flow to reduce the need to raise outside capital. Alternatively, it may not have the option to raise capital even if it wanted to due to insufficient proof-of-concept. Once the startup has become more established and made more progress, it can raise capital with more negotiating leverage (rather than being at the mercy of early investors).

For an early-stage company, what are a few options for liquidity events?

A liquidity event is an opportunity for the founder of a private company and its early investors to sell some or all of their equity ownership. Given the illiquid nature of the investment and the high-risk nature of early-stage companies, it's reasonable to see a liquidity event.

Often, this will begin with a minor partial liquidity event to take some profits (i.e., partial “cash-out”), with the provider of the capital being a venture capital firm. Later examples of liquidity events as the startup grows will be raising financing from venture firms and being acquired by a strategic.

Who are angel investors?

An angel investor is most often a wealthy, accredited individual investing in seed and early-stage companies. Once a startup has exhausted its funding from friends and family members, the angel investor is often the first outside investor to provide capital.

The company may often not even have a working product, and therefore, the angel investor is taking on a substantial risk of losing their initial investment. For this reason, the check size of an angel investor is rarely more than \$1 million per startup investment.

Walk me through the various funding rounds in venture capital.

Venture Funding Rounds

- **Seed Round:** The seed round will involve friends and family of the entrepreneurs and individual angel investors. Seed-stage VC firms can sometimes be involved, but this is typically only when the founder has previously had a successful exit in the past.
- **Series A:** The Series A round would include early-stage investors and typically represents the first time institutional investment firms will provide financing. During this stage, the startup's focus will be on optimizing its product offerings and business model and developing a better understanding of its users.
- **Series B/C:** These rounds represent the expansion stage and still involve mostly early-stage venture firms. The startup has gained initial traction and shown enough progress for the focus is now be trying to scale, which involves hiring more employees (especially in sales & marketing, business development).
- **Series D:** The Series D round (and onward) represents late-stage investments where the new investors providing capital will usually be growth equity shops that invest under the belief the company has a real chance at undergoing an IPO in the near term.

From the perspective of an entrepreneur, what are the pros and cons of raising outside capital?

For entrepreneurs, the primary reason for raising outside capital is often first to ensure the business has sufficient funds even to operate. This will particularly be the case for startups that require high R&D for continued product development and to scale further.

Using the newly raised capital, the entrepreneur has various options to improve their operations, including:

- Hire more talented employees and increase overall employee count (more job functions)
- Move the business to a better location (i.e., be closer to target customers, VCs, a state with lower taxes)
- Invest more in sales & marketing, advertising, and other operational roles
- Implement new growth strategies that otherwise could not be done without this funding

Most venture capitalists are rarely simple providers of capital. These investors are often former entrepreneurs who had their own successful exits and can develop a strong mentor-mentee relationship with the entrepreneur. Under the guidance of the new advisors, the entrepreneur can better navigate the difficulties all startups inevitably face as they attempt to achieve growth and scale.

In addition, an investor's willingness to not only risk the loss of their capital but to spend their time and resources to provide guidance to the management team proves the validity of the entrepreneur's business idea (i.e., this product and business plan could be viable).

The major downside of raising outside capital is that the entrepreneur must give up more ownership of their business as more capital is raised. However, this is often a necessary decision if the entrepreneur wants to grow the business and eventually IPO (or have a similarly successful exit).

What does the proof-of-concept stage involve?

When a company is at the proof-of-concept stage, there's no working product on-hand. Instead, there's just a proposed idea for a certain product, technology, or service. Thus, it's difficult to raise much capital; however, the amount of funding required is usually very minimal since it's only meant to build a prototype and see if this idea is feasible in terms of product-market fit. At this stage, the investors providing this type of seed investment are usually friends, family, or angel investors.

Explain to me what the commercialization stage is.

The commercialization stage is when the value proposition of a startup and the possibility of a product-market fit has been validated, meaning institutional investors have been sold on this idea and contributed more capital. The focus shifts almost entirely to revenue growth, as profitability is not the priority, since the company has more than enough capital raised and investors will contribute more if needed.

The commercialization stage usually refers to Series C to D (and beyond), and there'll be several large, institutional venture firms and growth equity firms involved in guiding this high-growth company to help refine the product or service offering, as well as the business model.

Explain the power law of returns in venture capital investing and its implications?

Peter Thiel famously said, "The biggest secret in VC is that the best investment in a successful fund outperforms the entire rest of the fund combined." The return distribution that Thiel was referring to is known as the power law of returns.

Venture capital investments are made under the expectation that most will inevitably fail. However, a single investment by itself can enable the firm to meet its return hurdle. The implication of this is the considerable emphasis on market size when looking for prospective companies given the risk/return profile.

Given the rate of failure in startups, VCs only target investments that have the potential to return the value of the entire fund.

If the revenue opportunity present in the market is not large enough, the fund could not meet its return threshold even if the startup turned out to be a success. This illustrates why VCs target large industries worth at least ~\$500 million to achieve their target of ~\$100 million revenue based on their market penetration assumptions with a reasonable margin of safety.

What role does a lead investor have in a round of financing?

The lead investor in a company is a venture capital firm or individual investor that organizes a round of funding for a particular company. The lead investor is usually one of the first institutional investors in the company and will contribute the largest amount of capital (i.e., “lead the round”).

The lead investor will often use their network to help the startup raise more funding from various investment firms and make introductions as needed. Therefore, the more credibility the lead investor has within the investment community, the higher the likelihood that other VC firms will trust their judgment and be more willing to participate in the upcoming financing rounds.

Give me an example of the drag-along provision in use?

The drag-along provision protects the interests of the majority shareholders (usually the early, lead investors) by enabling them to force major decisions such as exiting the investment.

This provision will prevent minority shareholders from holding back a particular decision or taking a specific action, just because a few shareholders with small stakes are opposed to it and refusing to do so.

For example, suppose the stakeholders with majority ownership desire to sell the company to a strategic, but a few minority investors refuse to follow along (i.e., drag-along the process). In that case, this provision allows the majority owners to override their refusal and proceed onward with the sale.

What are the typical characteristics of preferred stock?

Preferred stock can be best described as a hybrid between debt and equity – and sits right in between the two in the capital structure. Preferred stock has a higher claim on assets than common stock and receives dividends, which can be paid out as cash or “PIK.”

Unlike common equity, the preferred stock class doesn't have voting rights but has seniority in the capital structure, but still has a lower claim than any debt holders. Sometimes preferred stock can be convertible into common equity, creating additional dilution.

What are the two main types of preferred equity investments?

1. **Participating Preferred:** The investor receives the preferred proceeds (i.e., dividends) amount + has a claim to common equity afterward (i.e., “double-dip” in the exit proceeds).
2. **Convertible Preferred:** Also referred to as non-participating preferred, the investor receives either 1) the preferred proceeds or 2) common equity conversion amount – whichever is of greater value.

What is a liquidation preference?

The liquidation preference of an investment represents the amount the owner must be paid at exit (after secured debt, trade creditors, and other company obligations). The liquidation preference determines the relative distribution between the preferred shareholders and the common shareholders.

Often, the liquidation preference is expressed as a multiple of the initial investment (e.g., 1.0x, 1.5x).

$$\text{Liquidation Preference} = \text{Investment \$ Amount} \times \text{Liquidation Preference Multiple}$$

A preferred shareholder has a 2.0x liquidation preference. What does this mean?

A liquidation preference is a clause in a contract that gives a certain class of shareholders the right to be paid ahead of other shareholders in the event of a liquidation. This feature is commonly seen in venture capital investments. Given the high failure rate in venture capital, certain preferred investors desire assurance to get their invested capital back before any proceeds are distributed to common stockholders.

If an investor owns preferred stock with a 2.0x liquidation preference – this is the multiple on the amount invested for a specific funding round. Thus, if the investor had put in \$1 million with a 2.0x liquidation preference, the investor is guaranteed \$2 million back before common shareholders receive any proceeds.

What does it mean if the liquidation preference is a capped participation preference?

A capped participation preference (often called “capped participating preferred”) indicates that the preferred shareholders can share in the liquidation proceeds on a pro rata basis until total proceeds reach a certain multiple of the original investment, plus any accrued dividends. This is similar to participating preferred equity, but the proceeds are capped once a certain multiple has been reached.

What is the difference between the pre-money and post-money valuation?

- **Pre-Money Valuation:** The pre-money valuation refers to the company's value prior to the first (or the next) financing round.
- **Post-Money Valuation:** The post-money valuation accounts for the new investment amount after the financing round. It can be calculated as simply the new financing amount added to pre-money valuation, or the formula below can be used:

$$\text{Post Money Valuation} = \frac{\text{New Financing Amount}}{\% \text{ of Equity}}$$

Tell me about the difference between an up round vs. a down round.

Prior to a new financing round, the pre-money valuation will first be determined. The difference captured between the starting valuation and then the ending valuation after the new round of financing determines whether the financing was an “up round” or a “down round.”

- **Up Round:** An up round refers to financing in which the valuation of the company raising additional capital increases when compared to its prior valuation.
- **Down Round:** A down round refers to when the valuation of a company has decreased after a financing round when compared to the previous financing round.

Can you give me an example of when dilution would be beneficial for a founder?

As long as the startup's valuation has increased sufficiently (i.e., “up round”), dilution to the founder's ownership can be beneficial. For example, let's say that a founder owns 100% of a startup that's worth \$5 million. In its seed-stage round, the valuation was \$20 million, and a group of angel investors collectively want to own 20% of the company in total. The founder's stake will be reduced from 100% to 80%, while the value owned by the founder has increased from \$5 million to \$16 million post-financing despite the dilution.

Can a startup recover from a down round?

For certain, despite the dilution and potential internal conflict created from the unsuccessful new round of financing – the capital raised could have eliminated the risk of bankruptcy and given it enough money to turn the business around. This down round may have been the lifeline the startup needed to stay afloat and still have a chance at achieving its goals. While a down round is not a positive sign, it's not necessarily a sign that the end is near (although it may be more challenging to raise capital in the future unless clear improvements in performance are made).

What are anti-dilution protection clauses?

The purpose of anti-dilution protection clauses protects shareholders (usually the early investors) following a down round. In effect, their conversion ratio remains the same to shield their investment from potentially losing value due to the additional dilution created post-financing.

What is the term sheet in venture capital?

Term sheets establish the specific agreements of investment between an early-stage company and a venture firm. The term sheet is a non-binding agreement that forms the basis of more enduring and legally binding documents, such as the Stock Purchase Agreement (SPA) and Voting Agreement.

The term sheet is usually less than ~10 pages and is prepared by the investor. Although short-lived, the term sheet's purpose is to list out the initial specifics of an investment, such as the valuation, dollar amount raised, class of shares, investor rights, and investor protection clauses.

The term sheet would then flow into the capitalization table, which is essentially a numerical representation of the investor ownership specified in the term sheet.

What is the pay-to-play provision and what purpose does it serve?

A pay-to-play provision incentivizes investors to participate in future rounds of financing. Simply put, these provisions will require existing preferred investors to invest on a pro rata basis in subsequent financing rounds. If not, the investors will lose some (or all) of their preferential rights, which most often include liquidation preferences and anti-dilution protection. In most cases, the preferred shareholder will accept being automatically converted to common stock with a down round.

What is a right of first refusal (ROFR) and is it an interchangeable term with a co-sale agreement?

While a ROFR and co-sale agreement are both provisions intended to protect the interests of a certain group of stakeholders, the two terms are not synonymous.

- **Right of First Refusal:** The ROFR provision gives the company and/or the investor the option to purchase shares being sold by any shareholder before any other 3rd party.
- **Co-sale Agreement:** The co-sale agreement provides a group of shareholders the right to sell their shares when another group does so (and under the same conditions).

What are redemption rights?

A redemption right is a feature of preferred equity that enables the preferred investor to force the company to repurchase its shares after a specified period. It protects them from a situation when the company's prospects turn bleak. However, this is a rare occurrence to see exercised, as most of the time, the company would not have sufficient funds to make the purchase even if legally required to so.

What is a full ratchet provision, and how does it differ from a weighted average provision?

- **Full Ratchet Provision:** A full ratchet is an anti-dilution provision that protects early investors and their preferred ownership stakes in the case of a down-round, the investor with the full ratchet's conversion price will be re-priced to the lowest price at which any new preferred stock is issued. In effect, the investor's ownership stake is maintained at the expense of substantial dilution to the management team, employees, and all other investors.
- **Weighted Average:** A similar anti-dilution provision used far more often is the weighted average method, which uses a weighted average calculation that adjusts the conversion ratio to account for any past share issuance and the prices they were raised. In effect, the conversion rate is lower than that of a full-ratchet strategy, making the dilutive impact far less severe.

What is the difference between broad-based and narrow-based weighted average anti-dilution provisions?

Both broad-based and narrow-based weighted average anti-dilution protections will include common and preferred shares. However, broad-based will also include options, warrants, and shares reserved for purposes such as option pools for incentives. Since more dilutive impact from shares is included in the broad-based formula, the magnitude of the anti-dilution adjustment is thereby lower.

What is a capitalization table and what purpose does it serve to early-stage companies and their shareholders?

A capitalization table ("cap table") for any early-stage company tracks the equity ownership of a company in terms of number and type of shares (as well as series) along with any special terms such as liquidation preferences or protection clauses. For this reason, a cap table must be kept up-to-date to calculate the dilutive impact from each funding round, employee stock options, and issuances of new securities or convertible debt. Thus, all stakeholders can accurately calculate their share of the proceeds in a potential exit (i.e., liquidation event such as a sale to a strategic or IPO).

What is an option pool?

An option pool is a reservation of a certain percentage of shares (usually between 5% and 20%) set aside for future issuance to key employees, investors, or advisors of a company. The option pool is often used to attract talented employees by a cash-strapped startup that cannot afford to pay out high salaries due to insufficient cash and provide an incentive to perform well for the options to vest.

Another case an option pool is used is when a private equity firm performing an LBO allots a portion of equity as an incentive tool for the management team to reach their targets (called a "management option pool").

What is an employee stock ownership plan (ESOP) pool?

When a startup has an employee stock ownership plan (ESOP), it allocates a certain percentage of shares to early employees and later hires. The key distinction of this specific type of options pool is that it's not intended for the founding team, first employees, or current employees. Instead, an ESOP aligns incentives for future employees and is used to hire the top talent once the business scales. By the time a startup is nearing its Series A, the ESOP pool size typically ranges around ~10% of the fully diluted equity.

What is growth equity and how does it differ from earlier-stage venture capital investments?

Growth equity involves taking minority equity stakes into high-growth companies that have moved beyond the initial startup stages. The investments made by growth equity firms are often called growth capital because they help the company advance into the next development stage. Since the growth equity firm doesn't have a majority stake, the investor has less influence on the portfolio company's strategy and operations. However, the objective is more related to riding the ongoing, positive momentum and taking part in the IPO ideally.

How does the typical growth equity investment differ from traditional buyouts?

Traditional buyout funds take majority stakes in stable growth, mature companies (usually ~90-100% equity ownership), whereas growth equity investors take minority stakes in high-growth companies attempting to disrupt a particular industry. For growth-oriented investors, differentiation is a key factor that's often the leading rationale for investing. To these investors, what makes a product worth investing in is the value it derives from its proprietary technology that's difficult to replicate or protected by patents. The end goal in growth equity investing is almost always an IPO.

When growth equity investors pursue industries to invest in, how does it differ from industries targeted by traditional buyout firms?

Growth equity is centered on disruption in “winner-takes-all” industries; whereas traditional private equity is more about defensibility in revenue and free cash flows. In the industries that traditional buyouts take place, there's enough room for there to be multiple “winners.” In addition, traditional buyouts typically rely heavily on the use of leverage in their investments, in contrast to growth equity firms, which may not use any debt, and usually are more focused on the pure growth of the equity in their investments.

What is an initial public offering (IPO)?

An initial public offering (IPO) is the process of a privately held company offering its equity to the public in a new stock issuance. For a company to undergo an IPO, specific requirements must be met that are set by the Securities and Exchange Commission (SEC). An IPO presents a company with an opportunity to get further capital by offering shares through the primary market channel.

Traditionally, the company will hire several investment banks (often called the “underwriters”) to advise on the transaction as the company “goes public.” The investment bank will market the company (i.e., the “roadshows”), secure demand from institutional investors, and set the IPO price and listing date.

From the perspective of management, what are some downsides of going public?

While the benefits of becoming publicly traded from a monetary perspective are clear, there are several disadvantages to management once their company has gone public:

- The foremost concern for management teams is that they could lose control over their company, especially if shareholders are not pleased with its financial performance and believe they should be replaced.
- The disclosure requirements will open up the company to scrutiny by investors and their closest competitors can access their financials as their operations and playbook are an open-book for viewing.
- Going public puts more pressure on the management team to meet near-term performance metrics, as public market investors can be more short-term focused and expect companies to meet their quarterly or annual earnings guidance.

What are the key differences between common and preferred stock?

- **Common Stock:** Common stock is often referred to as basic shares and represents ownership in a particular company. Usually, common shares come with voting rights, and the amount of voting power a common shareholder has is directly related to the number of shares owned.
- **Preferred Stock:** Preferred stock is a hybrid between bonds and common stock, which can be appealing to more risk-averse investors. Preferred stockholders are usually paid preferred dividends consistently, hence its comparison to bonds. But if dividends cannot be paid out to any equity investors, preferred shareholders have a higher claim on the distributions than equity shareholders – so no dividends can legally be paid to common shareholders without first paying the preferred shareholders. On the downside, preferred shareholders have no (or limited) voting rights in the company's corporate governance and are still below all debt holders in the capital structure. In the event of liquidation, the preferred shareholders will have a higher claim on the liquidated assets relative to common shareholders, assuming there's remaining collateral after the debt holders have received their share.

What are preemptive rights?

Preemptive rights allow existing shareholders to purchase a new issuance before it's offered to other potential buyers. This provision protects early shareholders from dilution when the company issues new shares.

How can founders maintain control over their company post-IPO?

Once a company has gone public, it's inevitable for ownership to become diluted for all stakeholders. For many founders, this is the primary concern when considering selling shares to the open public.

Three methods for management to help protect their control over their company are:

1. **Create Different Share Classes (Dual-Class Share Structure):** This would mean there are distinct classes of common stock with different voting rights or voting power. The most common practice is the issuance of Class A and Class B shares. When a company goes public, the founders and early investors can be given shares with extra voting rights (often called "super-voting shares").
2. **Maintain Majority Control:** While it's near impossible for the founders to maintain majority control, close relationships with their early investors and investors with large stakes in the company would be very beneficial to have on their side and a way for them to maintain majority control in the company to an extent. Having the backing of the largest shareholders is reassuring from the perspective of management.
3. **Control the Internal Board of Directors (BOD):** While regulations are increasingly forcing companies to include more independent board members, the more BOD members that have a close relationship with the founders – the more beneficial it is for management. Ideally, the entire board respects and believes the current management team is the right team to lead the company. The absence of this type of dynamic could lead to management being voted out if performance suffers. Therefore, it's in management's best interest to have as many inside directors as possible (rather than outside directors).

What are no-vote common shares and which headline IPO brought it into discussion?

The mainstream IPO that included no-vote common shares was the IPO of Snap Inc. in 2017. While common shares with differing voting rights are frequently seen during IPOs, the no-vote common shares were a rarity. In Snap's IPO, most shareholders were not given voting rights, which was controversial, as investors would be completely hostage to management decisions under the proposed corporate governance. Even Snap's S-1 filing wrote that "to our knowledge, no other company has completed an initial public offering of non-voting stock on a US stock exchange" and acknowledged this decision could lead to a lower share price and fewer investors.

In Snap's IPO, there were three classes of stock: Class A, Class B, and Class C. Class A would be the shares traded on the NYSE with no voting rights, Class B would be for early investors and executives of the company and come with one vote each, and Class C would be held only by Snap's two co-founders, CEO Evan Spiegel and CTO Bobby Murphy. The Class C shares would come with ten votes apiece, and the two holders would have a combined 88.5% of Snap's total voting power post-IPO.

During an IPO, what is the role of the underwriter?

The underwriter, most often an investment bank, will serve as an intermediary between the company issuing securities and the investing public. The underwriter will be tasked with handling the new issuance of stock and to ensure the public, primarily institutional investors (e.g., mutual funds, pension funds, hedge funds), commit to purchasing the issuance before actually becoming available for purchase in the open market. Some key obligations that the underwriter will have are to handle the roadshow strategy, generate interest from potential investors, and price the IPO optimally to maximize the value received.

Read More → [Raising Capital & Security Underwriting](#)

What is the purpose of a syndicate of underwriters?

A group of investment banks, called a syndicate, will purchase the new issuance of securities for a negotiated price and then promote the securities to their network of investors in a process called a roadshow. Doing so spreads the risk across several underwriters instead of placing all the pressure on one underwriter, which maximizes the chance of the entire issuance being sold to the public (but mainly to institutional investors).

What events take place during management roadshows?

A roadshow is when the management of a company raising capital will travel around to give presentations to institutional investors such as large asset management firms, hedge funds, and pension funds. The reason for doing the roadshows is to generate interest from investors for an upcoming issuance of equity or debt.

What is the difference between the primary and secondary markets?

- **Primary Market:** The primary market is when securities are first issued via an IPO – i.e., this is the first time shares are offered to the public.
- **Secondary Market:** The secondary market is when the securities are traded amongst investors, which include institutional firms (e.g., asset managers, hedge funds), as well as individual investors.

What is a red herring, and when is it filed with SEC?

More formally called the preliminary prospectus, the red herring provides prospective investors with information regarding an upcoming IPO. Included in the prospectus are details related to the company's products, growth plans, intended uses of the raised funds, potential risk factors, and more background details on the company (e.g., detailed financial statements, management team biographies).

The filing must be filed with the SEC, not just accompany the bankers on the roadshow and be used to raise investor interest, but to describe the issuance of equity and the proposed details of the IPO offering.

What is the S-1 filing, and how does it differ from the red herring?

The SEC Form S-1 is a required filing with the SEC for all US companies seeking to become publicly listed and registered on an exchange (i.e., planned IPO). Found in the S-1 is information on the planned usage of the raised capital proceeds, the current business model, full detailed historical financials, commentary on the competition landscape, a prospectus of the planned security issuance, the method for setting the offering share price, and commentary on existing securities and dilution.

Compared to the red herring, the S-1 is lengthier and more formalized regarding the IPO's information and set strategy. The red herring is a preliminary prospectus that comes before the S-1 and is circulated during the "quiet period" before the registration has become official with the SEC. Often, the SEC may request additional material to be included or changes to the red herring. The final prospectus, the official S-1, would then be filed before the company can proceed with the distribution of shares.

Why is there a lockup period associated with IPOs?

The 180-day lockup period associated with IPOs prevents insiders from selling their shares immediately once the company has undergone an IPO. The sudden increase in selling volume could cause the share price to drop significantly, as many early investors might want to liquidate their positions as soon as possible.

What is the difference between a seasoned equity offering (SEO) and a rights offering?

- **Seasoned Equity Offering:** A SEO is when a company that's already public (i.e., underwent an IPO) issues additional shares. This is often referred to as a "follow-on offering."
- **Rights Offering:** A rights offering gives existing shareholders the exclusive right to purchase new shares at a pre-specified price below the current share price on a pro rata basis.

During an IPO, what does a firm commitment mean?

A firm commitment refers to when an investment bank, the underwriter of an IPO, agrees to assume all the risks associated with failing to sell all the shares being listed by purchasing all the securities being issued directly from the company. Under this method, the underwriter would bear all the risk because it purchases the entire issuance of new securities (hence, syndicates are far more common).

What do “best efforts” underwriting agreements involve?

An underwriting arrangement based on “best efforts” refers to the equity underwriters' commitment to making their best effort to sell as much as possible of the securities issuance. However, there's no guarantee that all the securities will be sold, so there are no negative implications for them failing to achieve the set capital raising target (other than potential reputational damage).

Could you explain the Dutch auction pricing method in an IPO?

In an IPO that uses the so-called Dutch auction pricing strategy, the share price is lowered until all the shares would be sold. Then, all the shares are sold at that price, which is the highest price where all the shares would be sold. Rather than setting the price based on valuation alone, the price is based on the bids placed by the interested buyers that will submit how much they would pay and the number of shares they can buy at their proposed price. In effect, bringing in a game theory element to this type of auction pricing strategy.

What is a direct listing and how does it differ from the traditional IPO?

An alternative to the traditional IPO is a direct listing, which involves the company selling shares directly to the public market without the advising of an investment bank. Companies that decide to proceed with this route can avoid the lock-up period associated with IPOs, minimize the dilutive impact, and save a significant amount of money by not having to pay fees to investment banks. This option is riskier, however, since the process is relatively new, and there's no assurance that the shares will be priced correctly or enough shares will be sold.

Direct listings come with fewer transaction costs (paid to investment banks) and the ability to avoid the lockup period but are not viewed as a “replacement” for an IPO because they're not ideal for a company that wants to raise capital. However, the SEC recently announced a rule change that companies undergoing a direct listing can now raise capital, making direct listing a more compelling alternative to IPOs. One of the first companies to do a direct listing was Spotify in 2018, as well as Slack in 2019. Two of the most prominent IPOs in 2020, Palantir Technologies and Asana Inc, both opted to go public under the non-traditional direct listing.

In the past couple of years, IPOs have increasingly been scrutinized for leaving “money on the table.” Could you explain what this criticism is regarding?

Following an IPO, the shares often surge and the market capitalization of the company can jump in excess of \$50-\$80 million on the 1st day of trading. This has led to criticism, including from prominent VC Bill Gurley, one of the leading critics of the underpricing of IPOs by investment banks for “leaving money on the table.”

In defense of investment banks, this underpricing can be attributed to how their priority is to sell all the shares being listed, which often requires pricing it lower than some would pay. However, many investment banks are still accused of having conflicts of interest to serve both the company going public and institutional investors. For example, imagine an unrealistic scenario where shares were priced to “perfection” and the share price didn't budge once listed, meaning no share appreciation for the investors. Therefore, this discounted feature of IPO pricing could be seen as a way to establish long-term relationships with investors because if they achieve a high return from an IPO, they'll be more likely to invest in the next IPO managed by the same investment bank.

A syndicate in an IPO has exercised the greenshoe option. What does this mean?

The greenshoe option (or “over-allotment”) is a clause outlined in the underwriting agreement of an IPO that allows the underwriting syndicate to purchase up to an additional 15% of the shares at the offering price. The option can be exercised, and more shares can be sold if demand from the public exceeded expectations, and the shares are trading above the offering price, which allows the issuing company to raise more capital. Alternatively, the underwriters are granted this right to buy more shares from investors to stabilize the price and decrease the supply if the share price dropped.

What is a special purpose acquisition company (SPAC)?

A special purpose acquisition company (SPAC), often referred to as a “blank check company,” is a shell company formed strictly to raise capital through an initial public offering to acquire an existing company or merge multiple companies. Prior to having a specific target, the SPAC raises funds through an IPO of the SPAC’s equity securities, with the sponsor retaining ~20% of the post-IPO SPAC. The invested capital is moved into an escrow until the target to be acquired is determined, or else the capital will be returned to investors. While SPACs have been around for a while, 2020 has been a record year for this type of investment vehicle.

From the company’s perspective, SPACs have quickly become a popular method to go public in a simple, time-efficient manner (usually ~4-6 months). Arguably, it’s even more convenient than a reverse merger with added benefits such as involving significantly less work than the traditional IPO. Most of the strenuous work involved in an IPO, such as making investors interested and raising capital, has already been done by the SPAC sponsor. The capital has already been raised, so all that remains is for the company to agree to a valuation. Here, the valuations paid technically have more accuracy (or at least are more straight-forward) since a SPAC deal is more like an M&A negotiation with buyers rather than an underwriter telling you what the market might pay. Thus, SPAC acquisitions come with a certainty of pricing aspect and less risk of “leaving money on the table” because there’s no risk of failing to issue all the shares (hence no need for a discount).

SPACs provide many advantages for companies looking to go public compared to a traditional IPO or direct listing, such as simplicity, faster timeframe, better valuation, and well incentivized long-term partners. As a side benefit, most SPAC sponsors are reputable individuals and asset management firms; otherwise, they would not have been able to secure the capital in the first place.

Read More → [Going Public Circa 2020; Door #3: The SPAC](#)

What is a private placement?

A private placement is when a company raising capital issues shares to only a select few institutional investors. In contrast to a public issuance, the securities are not made available for sale to the open market and are not required to be registered with the SEC. A private placement is a more direct approach to selling to a group of accredited investors, consisting of institutional firms, mutual funds, and pension funds. Both public and privately held companies can partake in private placements.

What are the characteristics of a shelf offering?

Filed with the SEC, a shelf offering provision (S-3 registration) enables a company to make an issuance in a three-year period, rather than selling it all in one sale. This can be beneficial as the market conditions could turn more favorable, and it gives the company raising capital more flexibility with the timing of the sales. Hence, many shares will be “on the shelf” until the company sells them.

What does a private investment in public equity (PIPE) mean?

Private investment in public equity (PIPE) refers to the private placement of securities in a public company by accredited investors such as hedge funds, private equity funds, and mutual funds. The purchase will be made at a discounted price, usually as an unregistered convertible or preferred security.

PIPEs have become an alternative way for companies to raise capital and this financing structure became more prevalent due to the relative cheapness and efficiency vs. a traditional secondary offering. There are also fewer regulatory requirements, as there’s no need for expensive roadshows.

General Market Knowledge

Questions pertaining to the public equity markets tend to be less common in investment banking and private equity interviews when compared to hedge fund, sales & trading, equity research, and capital markets roles. You should still have a general understanding of the markets as a baseline.

If you have not already, make it a daily habit to read the [Wall Street Journal](#). Most university career centers will provide students with free access. If not, WSJ is currently offering a \$4 per month student digital pack offer on their website as of right now. Even if this current offer ends, WSJ is known for giving significant discounts to students.

Also subscribe to email newsletters such as [Morning Brew](#) and [Dan Primack's Pro-rata](#) – these are two convenient ways to keep tabs on what's going on in the markets.

What did the S&P 500/Dow/NASDAQ close at last night?

The market would likely be open at the time of the interview, and the interviewer doesn't expect you to have the exact latest numbers. Therefore, saying: "The S&P 500 closed around 3,570 yesterday and opened lower this morning after reports of new coronavirus infections surged" would be sufficient. Instead, it's more important to discuss the recent trends and developments driving equity valuations over the past few months and years rather than memorizing the precise figure at which a certain index closed.

If you're preparing for an interview during an important economic period or event (e.g., COVID-19, recent Fed meeting, financial crisis, healthcare reform, tax reform), be aware of what's going on and be prepared for questions on your thoughts on those types of current events.

What is the current 10-year risk-free rate?

Similar to the previous question, you just need to answer with an approximation of the current risk-free rate. A response along the lines of "The yield on the 10-year treasury bonds has been around ~0.8% as of late" will do. Just make sure you're within the ballpark. If you want to open up a more in-depth discussion about the risk-free rate and interest rates, you could mention the direction it has been trending. For example, you could mention that throughout the fall, the risk-free rate has been trending upward since the summer from the increased political risks surrounding the 2020 elections, the rising cases of coronavirus cases as more businesses re-open, and more questions surrounding the timeline of the vaccine development.

What's your near-term and long-term outlook on the equity markets?

Start by answering where the current level of the market is and the prevalent trends in the market. For instance, has the market been rising with strong earnings? Then talk about the key catalysts that could move the market and the specific development you would be looking out for.

In addition, try to find out what the bank's research analysts predict for the market's Year-End Target. You can find research estimates on Bloomberg or ask an analyst you have networked with to share the latest estimates with you. Many brokerage firms also provide institutional research to retail clients using their online brokerage account, so opening up a Chase or a Merrill account can allow you access to J.P. Morgan and BofA research.

Later on in your careers, when you discuss the markets with clients, you'll start with your house view (research), and then you can add your personal views on why you would disagree.

Sample Response

"First off, I know your bank's Year-End target for the S&P500 is 4,000. I am personally slightly more bearish with the view that the S&P 500 will trade range bound and close the year at the same point we are now. Overall, the S&P 500 is trading at a P/E ratio of 36, which is quite high on a historical context. Most of the positive news on the COVID recovery has already been priced in, and I feel a shift in sentiment or flows can send stocks lower. We are approaching quarter-end, and with stocks outperforming bonds this quarter, I would expect some investors to sell stocks and buy bonds for rebalancing.

Although I am slightly more bearish, I don't think there's a risk for a broad market crash. The Fed has shown strong support for the market. We are seeing the Fed ease up on some support in the market. The S&P 500 is already higher than where we were pre-COVID, and my view is that the Fed will continue to remove support for the market if we move higher, limiting the upside gains in the market and causing us to trade on a range-bound basis."

What do you think the Fed is going to do at the next FOMC Meeting?

Although the question asked has no right or wrong answer, there are a few unasked questions that will showcase your understanding of the markets:

- When is the next FOMC meeting?
- What is the current Fed Funds Target Rate?
- What was the Fed's action at the last FOMC meeting?
- What is the Futures Market implying for the FOMC meeting?

To answer this question effectively, you need to list the key facts to the unanswered questions and then express your view. If the market is not expecting a move in the Fed Funds rate, be prepared to discuss the Fed's balance sheet and changes in its Asset Purchase Programs.

Sample Response

"The next FOMC meeting will be on November 5th. The current Fed Funds Target rate is between 0 and 25 basis points. The Fed has held this rate steady since the emergency meeting in March when the Coronavirus crisis began. The market is not expecting any change to the Fed Funds Target Rate, and I'm not either.

I'll be watching the Fed closely to see if they're planning to change the language or guidance of their asset purchase program. With the stock markets near all-time highs, I would expect the Fed to remove some asset purchases modestly they have promised but gradually as to not spook the markets."

What is the key difference between the NYSE and the NASDAQ?

In the US, the two primary stock exchanges are the NYSE and the NASDAQ. Before filing for an IPO, companies must decide which stock exchange to list their shares.

- **NYSE:** The NYSE (New York Stock Exchange) is the older stock exchange with roots back to 1817 and physically on Wall Street. Originally, the NYSE was setup with pit trading on an actual trading floor, but today, most trades are electronic, although some market makers still exist.
- **NASDAQ:** Launched in 1971, the NASDAQ is newer and a completely electronic exchange with offices in Times Square and data centers in New Jersey. Compared to the NYSE, the NASDAQ is more tech-heavy due to its flexible listing requirements that appeal to former startups that at one point didn't qualify for the NYSE. However, the NYSE has relaxed its listing requirements in recent years, removing the clear divide as there used to be.

What is the treasury yield curve, and what does its shape tell you?

The yield curve is just a plot of the relation between the yield and the term of otherwise similar bonds. The treasury yield curve plots treasury bond yields across their terms and is the most widely used yield curve as it sets a “risk-free” benchmark for other bonds (corporate, municipal, etc.).

A treasury yield curve is normally upward sloping because, all else being equal, an investor would prefer a one year, 3% bond than being locked into a 30-year, 3% bond.

However, investors’ expectations about future interest rates affect the slope as well. When investors expect the Fed to raise short-term interest rates in the future, investors will decrease demand for short-term treasuries to avoid getting locked into a low but soon-to-be higher short-term rate. This decrease in the relative demand for short-term treasuries will raise short-term yields and flatten the yield curve.

What does an inverted yield curve tell you?

An inverted yield curve means that yields on longer maturities are lower than shorter maturities of otherwise comparable bonds, like treasuries. Normally, yield curves are upward sloping as issuers must pay a premium to entice investors to keep their capital locked up for a longer-term. When the yield curve inverts, it's usually a harbinger of an economic slowdown and recession. In fact, the last seven recessions were preceded by an inversion of the yield curve, and it's considered a strong leading indicator by economists and investors alike.

The inversion happens as investors anticipate market interest rates to decline down the road (presumably because they expect a slowdown and expect monetary policymakers to eventually lower rates to stimulate the soon-to-be slowing economy). Thus, investors prefer the safety of long-term maturities at the current higher rates over investing in shorter maturities and having to reinvest at the lower expected future rates.

In terms of yield curve jargon, what does an upward sloping yield curve, inverted yield curve, steeping yield curve, and flattening yield curve imply?

- **Upward Sloping:** Long-Term Government Bond Yields > Short-Term Government Bonds Yields
- **Inverted:** Long-Term Government Bond Yields < Short-Term Government Bonds Yields
- **Steepening:** The interest rate differential between Short-Term and Long-Term bonds is increasing
- **Flattening:** The interest rate differential between Short-Term and Long-Term bonds is decreasing

During a period of record unemployment and economic turmoil in 2020, the markets recovered faster and performed better than expected. What led to this disconnect between the equity market and the economy?

While there are many potential reasons the markets are trading at the levels they’re trading at today (or counter-arguments on whether they should be), one reason valuations have recovered rather quickly is because of the role that the risk-free rate has when valuing equities. Since the market bottomed in March 2020, the US 10-year treasury notes have been around 0.6% to 0.8% – mostly on the lower end of this range, especially between July and August 2020. Many institutional investors had two choices to either invest in:

1. Default-free government bonds yielding practically nothing (< 1% return)
2. Invest in the stock of large, multinational companies with a high likelihood of making it through this period, if not benefit from the lock-downs

The strong performance of the so-called “FAANG” stocks (Facebook, Amazon, Apple, Netflix, and Google) made it clear which decision many investors chose. These leading tech-giants have disproportionately outperformed the broader market as they led its recovery. Given their weighting in the S&P 500 Index, it becomes clear why the market surpassed record all-time highs later in 2020.

Most investors will either directly or implicitly make an investment decision based on the opportunity cost elsewhere. In the direct case, many investors will use discount rates that use the current ten-year government yield as the risk-free rate while performing a DCF (which leads to higher valuations). Others will implicitly see how bond prices have increased as more investors have sought safer assets, which means lower yields because of the inverse relationship. So investors who seek high returns (or even at least have the potential to receive a moderate return) have nowhere else to park their capital other than the equity market.

How does the current market compare to the peak of the Dot-com bubble in 2000?

The Dot-com bubble (often called the internet or tech bubble) reached its highest point around March 2000. At that period, the Chairman of the Federal Reserve, Alan Greenspan, announced his intention to raise interest rates aggressively. The US 10-year treasury notes were yielding ~6% on average. Therefore, when concerns about equities being overvalued were increasingly spreading – investors could sell their shares and receive a ~6% risk-free return.

Contrast this to today, where the Fed Chairman, Jerome Powell, has clarified interest rates will remain near zero through 2022. In the current market, there's no “safe-haven” investment with a decent yield. The term “zero-default risk” loses its meaning to equity investors when interest rates are close to zero (i.e., fair to state, it becomes almost worthless to investors seeking decent returns). The Fed has made it clear that interest rates will remain at this low rate until they're 100% confident that the economy has recovered from the coronavirus's negative effects. Despite providing projections that GDP will bounce back in 2021 and confidence in the recovery, there was no public signal that a rise in interest rates was even in consideration.

What is the definition of VIX and why is it used?

VIX is an index of the implied volatility of the one-month S&P 500 options at different strikes. VIX is a measure of overall market sentiment and is often referred to as the “fear gauge.” A low VIX is associated with a strong market with limited volatility. A high VIX is a sign of fear and an uncertain market. The VIX measures S&P 500 options' implied volatility, including how costly options are to hedge your portfolio.

Where is the price of oil at?

Be careful not to quote the futures prices, as both Bloomberg and Yahoo Finance will show futures prices as well as the price of the spot index. For Oil indices, be specific if you're quoting WTI or Brent, as there's more than one benchmark.

What tools does the US government have to combat inflation or hyperinflation?

The Federal Open Market Committee (FOMC) targets an inflation rate of ~2% over the long-term. To keep inflation around this target range, the US government has monetary and fiscal policies it can use to maintain a stable, long-term inflation rate.

- **Monetary Policies:** Monetary policies involve actions taken by the US central bank to control the money supply and demand to maintain sustainable economic growth. These policies can be classified as either expansionary or contractionary. Expansionary policies increase the liquidity of the money supply, whereas contractionary policy decreases the liquidity of the money supply. Examples of these policies would include strategically adjusting interest rates, changing reserve requirements, direct lending to banks, and open market operations.
- **Fiscal Policies:** Fiscal policies are related to changing tax policies and government spending to increase or decrease the purchasing power of the economy. In effect, this gives the Fed the ability to regulate the economy and influence economic conditions. For example, the corporate tax cut in 2017 from 35% to 21% under the Tax Cuts and Jobs Act (TCJA) was enacted to fuel economic growth and incentivized corporations to return their foreign operations to create more jobs in America.

What is quantitative easing (QE)?

Quantitative easing (QE) is a form of monetary policy where the central bank will directly make purchases of longer-term securities from the open market to increase the money supply and total liquidity. In effect, QE will decrease interest rates, which will directly encourage more lending and make money flow more into equities since fixed-income instruments will have low yields. This type of monetary policy is controversial, as it's essentially meant to flood the economy with money to spur economic activity. QE became a controversial topic of discussion when the Federal Reserve announced in March 2020 that it would purchase \$700 billion worth of government debt, bonds, and mortgage-backed securities over the coming year. This unprecedented decision would not only significantly increase the Fed's balance sheet risk and increase its already-mounting debt, but has created widespread concerns about this seemingly endless "printing" of money. The broader implications this could have for future generations remains an unknown. Often, to help spark economic growth, the Fed will turn to cut interest rates, but interest rates were already close to zero – hence, the Fed turned to QE.

What does it mean when the Fed is engaging in open market operations?

Open market operations are when the central bank purchases or sells short-term government securities (e.g., treasuries) in the open market to influence the money supply and impact short-term interest rates. If the money supply circulating in the system is increased, it becomes easier for businesses and consumers to get loans at low-interest rates, which has a positive impact on economic activity. Alternatively, the selling of securities from the central bank's balance sheet will remove money from the system. This reduction makes loans more difficult to be approved for (i.e., more demand from borrowers, less supply of money) and, as a result, makes interest rates on these loans increase.

What is the relationship between bond prices and interest rates?

There is an inverse relationship between bond price and interest rates. If market interest rates go up (due to a Fed interest rate hike or other market conditions), this has the impact of making the interest being offered by currently outstanding debt seem less attractive given the availability of comparable investments with now higher interest rates, all else being equal. The price of outstanding bonds would thus drop in this scenario.

If the Fed wants to lower rates, what can it do?

If the Fed wants to lower rates, it can directly purchase treasuries in the open market, and then T-bill yields would directly decline. The buyers, which consist of other foreign governments, individuals, institutional investors, and large banks, will store this cash and raise the supply. Alternatively, the Fed can lower rates through repurchase agreements (RPs). This would lower banks' demand for reserves and thus the Fed funds rate, which is the basis on which commercial banks lend.

If the Fed wants to increase rates, what can it do?

Conversely, if the Fed wants to raise rates, it can sell treasury bills in the open market. The Fed lends cash overnight to a select group of banks ("primary dealers") while taking treasuries as collateral at the repo rate. These primary dealers will then disseminate this cash throughout the financial system; thus, the fed funds rate increases. If the Fed wants higher rates, it can also do a reverse repo ("RRP"), in which the Fed borrows money from primary dealers. This would tighten the credit markets, and the fed funds rate would increase.

Besides direct purchases or sales, what is another policy tool the Fed can use to manipulate interest rates?

Another policy tool available to the Fed is to adjust the provided interest on bank reserves. The lowering of this rate (called interest on excess reserves, or "IOER") would encourage lending (and vice versa). The IOER is the interest rate the Federal Reserve pays on excess reserves. This rate is typically within the fed funds' target range. For example, if the fed funds target range is 1.5% to 1.75%, the IOER would be ~1.6%.

What is a repurchase agreement (repo)?

Repos (RPs) are short-hand for repurchase agreements (formally called “sale and repurchase agreements”) and refer to when a borrower, usually a government securities dealer, gets short-term funding by selling securities such as treasuries and agency mortgage securities to a lender.

The lenders could be money market funds, governments, and financial institutions. For a pre-determined period, the borrower will purchase the securities back for the original price plus interest (e.g., the repo rate), usually completed overnight.

What are reverse repurchase agreements (reverse repo)?

Conversely, a reverse repurchase agreement (reverse repo, or “RRP”) is when the purchaser of the security agrees to re-sell the security back to the seller for a pre-determined price at a later date. Essentially, repos and reverse repos are the two sides of a lending agreement from the perspective of the two opposing parties. Note, these are short-dated transactions with a guarantee of repurchase.

What is the Federal Open Market Committee (FOMC)?

The Federal Open Market Committee (“FOMC”) is the monetary policy-making body of the Federal Reserve System. The members meet eight times a year to set the interest rate policy and the federal funds rate. Following these meetings, a statement will be publicly issued to communicate the following changes in rates and commentary on their outlook on the economy and guidance on their forward-view of current policies (i.e., the potential for changes, risks considerations).

What is the federal funds rate?

The federal funds rate represents the target interest rate set by the Federal Open Market Committee (FOMC) for short-term loans. The fed funds rate target has historically been one rate (e.g., 1.5%) but is now a range (1.5% to 1.75%). Based on the agreed-upon Federal funds rate, commercial banks and institutions will borrow and lend their excess reserves to one other (usually completed overnight) at this rate. This is the rate that banks charge other banks for overnight loans to cover their reserve requirements. Banks with excess reserves can lend to banks with insufficient reserves. The federal funds rate affects the lending market because banks and many institutional lenders tie their lending rates (e.g., Prime, LIBOR) to the fed funds rate. As a result, any adjustments will flow throughout the entire market, including short-term US treasuries.

What is the federal discount rate?

The federal discount rate refers to the interest rate that the Federal Reserve charges commercial banks and other financial intuitions for short-term loans through the discount window loan process. In contrast to the federal funds rate, the federal discount rate is the rate charged to banks to borrow funds from them, whereas the federal funds rate is the rate that banks lend to each other. Compared to the fed funds target rate, the federal discount rate is typically 50 bps higher.

What are the reserve requirements set by the Fed?

The reserve requirement is the minimum level of cash a commercial bank is required to keep on-hand to cover its deposits. If this requirement were to be lowered by the Fed, the result would be more loans could be made. Therefore, lowering the reserve requirement is an expansionary policy option.

What would happen if interest rates turned negative?

In an environment with negative interest rates, the payment of interest flips between the borrower and lender. The borrower is being paid interest by the lender, rather than the other way around. Central banks use this drastic monetary policy tool to incentivize commercial bank lenders to make loans rather than sit on the funds.

From shortest to longest maturity, list treasury notes, treasury bonds, and treasury bills.

- 1. Treasury Bills:** T-bills mature within one year of issuance
- 2. Treasury Notes:** T-notes mature between two and ten years
- 3. Treasury Bonds:** T-bonds mature in twenty to thirty years

What is a certificate of deposit (CD)?

A certificate of deposit (CD) is an agreement with a bank to deposit money for a specified rate of return (i.e., earn interest). CDs are short-term agreements, issued at par based on the deposit amount, and have the same credit risk as a deposit.

What is the money market yield?

The money market yield is defined as the interest rate earned from holding securities of higher liquidity and short maturities (< one year). Examples of securities included would be certificates of deposit (CDs), US Treasury bills, and municipal bonds.

$$\text{Money Market Yield} = \left[\frac{(\text{Final} - \text{Initial Cash Flow})}{(\text{Initial Cash Flow})} \right] \times \left(\frac{360}{\text{Actual Days}} \right)$$

What are treasury inflated protected securities (TIPS)?

A treasury inflated protected securities ("TIPS") are issued by the US Treasury and protect bondholders from the risk of inflation. Periodically, TIPS adjust the par value (and the coupon) by the inflation rate for the bondholder to receive the originally expected return. Based on changes in inflation, the principal amount will be adjusted, which would then affect the coupon paid.

When does a liquidity trap occur?

A liquidity trap refers to when short-term interest rates are close to zero, which leaves the central bank unable to lower them further to stimulate economic activity. In effect, a liquidity trap makes monetary policy ineffective and restricts the options available to the Fed. Because of the near-zero interest rates, consumers hoard cash rather than invest it into bonds under the belief that the economy is headed in a negative trajectory.

What does the consumer price index (CPI) measure?

The consumer price index (CPI) is used to measure the change in prices that consumers pay for goods and services on average over a period. The index considers the prices of consumer goods (e.g., housing, apparel, transportation, education, recreation, food, medical care). CPI is followed by many to measure inflation and as a proxy to assess how effective (or not effective) the government's current economic policies are.

Would you agree with the statement that deflation is worse than inflation?

Deflation is the macroeconomic condition when a country experiences decreased prices. This is the opposite of inflation, which is characterized by an increase in prices. While initially, consumers may enjoy the lower prices, this cuts into the profit margins of companies. Then begins the cycle of employee layoffs, reduced pay, cutbacks on spending by companies, and overall, the output of the economy contracts, and the level of unemployment increases. This deflationary spiral will eventually lead to reduced investments made by companies to conserve cash, less overall economic growth, and increases in bankruptcies as borrowers cannot pay back loans.

From many economists' perspective, deflation is more serious than inflation as deflation is more difficult to control. An example of deflation's long-lasting effects can be seen in Japan, where even after decades of near-zero interest rates and quantitative easing (QE) efforts, the country has still failed to turn around and struggles to get prices rising again.

Hedge Fund Investing Strategies

Interviewing with a hedge fund is almost entirely dependent on pitching stocks and discussing investment trends in the market. The stock pitch you give is arguably the most important component of the interview. Regardless of whether you're an undergraduate/MBA student or coming from a stint in investment banking, all candidates need to have a stock pitch ready.

Just like other front-office finance roles, credentials and connections matter in the hedge fund industry. However, your investment thought process and differentiation in terms of the investment pitch you present are significant factors in the hedge fund industry.

Pitch me a stock.

Being asked to pitch a stock is a test of your understanding of how equities are valued and how markets function, as well as your ability to think about a business and industry. This is arguably the most important portion of a hedge fund interview as it opens up a lengthy discussion.

Come prepared to discuss one company in-detail and to defend your viewpoint of why its share price will increase (or decrease if pitching a short).

Before the interview, try your best to understand the firm's investment strategy and put together a pitch based on the information you can compile. This will require some creativity and guesswork, but it's very important to have a general sense of the fund's strategy heading into the interview.

It can be difficult to gather information on hedge funds, as most will not even have a website, or their website will show their logo and contact information due to regulatory reasons. However, you can still find bits-and-pieces of information to understand their focus vaguely.

In addition to being aware of the fund's strategy, avoid well-known stocks where the interviewer may have a predisposed view. For example, if you're a firm believer in the growth story of Tesla, but your interviewer is not, it's unlikely you'll change their mind. The interviewer will not wait three months and evaluate your performance vs. your peers. Therefore, it's recommended to pitch a lesser-known stock the chance the interviewer has an existing thesis on is rather unlikely. Even if they agree with your pitch, it'll lead to a far more granular discussion since they've looked at this company before.

Throughout your pitch, have conviction in your thesis and be prepared to hold your ground, as your assumptions will be challenged. No matter what, avoid changing your recommendation or even give off the impression of beginning to show doubt amidst giving your pitch. If your recommendation can be dismantled within a ten-minute discussion, you have either chosen the wrong stock to pitch or you didn't come adequately prepared to defend your assumptions.

Remember, counter-arguments with valid concerns can be made against all pitches. But as long as the assumptions that serve as the basis for your pitch remain intact, there's no reason to question the validity of your pitch – as that's precisely what the interviewer is attempting to do.

Understand which specific factors matter and which ones don't matter in the share price of the company you're pitching. When you can distinguish between the two and make it clear to the interviewer, the discussion becomes structured around those key points your pitch was crafted around – putting you in a far more advantageous position.

Stock Pitch Structure	
Recommendation	The pitch should begin with a clear statement on whether you recommend the fund to take either a long or short position. This introduction is your “hook” to grab the interviewer’s interest and your view on the potential upside (or downside) of an intriguing investment, so it’s important you get this part right to get the interviewer interested in hearing more about your investment thought process and rationale.
Company Overview	Next, contextualize the company’s “story,” which should progressively lead to the reason this stock you have chosen will perform better than its peers. Included in this section should be a basic overview of the business model with commentary on each of the segments, products/services sold, and any recent news that could affect the future performance of the business (e.g., M&A, management change).
Competitive Advantage	Then, explain what competitive advantage (i.e., economic moat) this company currently possesses. This section will be most relevant if you pitch a long position, but regardless of the fund strategy, try to show how you understand the competitive dynamics within an industry and focus on what differentiates this business that’ll protect its profits. Examples could be patents, intangibles such as branding, switching costs, or offering a product/service that cannot be easily replicated.
Market/Competitive Landscape	Next, the ongoing trends in the market that affect the company’s performance should be brought up, as well as your thoughts on the company’s current positioning within the competitive landscape. An approximation of the company’s market share and whether it’s a market leader or challenger should also be made clear.
Catalysts	Since the company’s background and market overview is done, the catalysts that can cause the market to “correct” its mispricing can be identified. Phrased another way: “Which specific catalyst will cause the share price to change?”
Price Target Range	In the next part, state an approximate price target range, how it compares to the current share price on a percentage basis, and the expected time for the target to be reached given the catalysts mentioned. While you need to mention the methods used to arrive at the valuation, a broad overview is sufficient because a significant portion of the follow-up discussion will be related to your valuation.
Investment Risks	Considering the interviewer has a deep understanding of the different risks specific to particular industries and the recurring risks that impact all industries, expect questions on the risk factors (and how they could be mitigated). If you’re interviewing with a fundamentals-oriented fund, you could bring up the margin of safety, consistent FCF generation in past years, low cyclicalities, and revenue diversification to defend how it could uphold during a recession.
Closing	Finally, close out the initial remarks by reiterating your earlier statement on the upside (or downside) potential. Then, expect an in-depth discussion with the interviewer on your price target, specific assumptions used for the DCF and comps analysis, and any points of disagreement/concern – which collectively measure how well you can defend your thesis using sound logic and reasoning.

Sample Fundamental Initial Pitch

"The company that I've chosen to pitch a long-position on is Genasys (NASDAQ: GNSS), a provider of mission-critical communications and emergency management solutions.

Genasys has three business segments, with the first segment being its Public Warning System (PWS) software, a location-based messaging platform for emergency messages sent out via mobile devices. This software benefits from the trend of critical communications solutions, and its use case is comparable to AMBER Alert. In 2019, the EU mandated its member nations to deploy country-wide PWS by June 2022 (Regulation Article 110 of the EEC), which should serve as a secular tailwind for Genasys.

Next, Genasys offers Long Range Acoustic Device (LRAD) and Acoustic Hailing Devices (AHD), which are types of speaker systems that provide long-range audio sirens to bring attention to an emergency or hazard in real-time. This segment is a legacy business, and industry growth has been on the lower end, but there are very few competitors of comparable size, and Genasys appears to have carved a defensible niche. Most of the customers for these products are specialized for the defense, law enforcement, and the military end markets, which fluctuate based on governmental budgets and depend on securing contracts. However, Genasys is becoming the global leader for LRAD systems, as shown by a recent \$14.5mm follow-on AHD order with the US Army and a \$4.3mm IDIQ contract with the US Navy.

Genasys' third segment offers Integrated Solutions, which combines hardware and software in a single system and represents the company's greatest growth potential, as it has a significantly larger market to sell. Large enterprises are signing contracts now, as shown by the recent telecom carrier contracts announced in Australia. In terms of M&A, Genasys made a complementary acquisition of the Ottawa-based emergency communications company, Amika Mobile. This continued geographic expansion through M&A, the long-term customer contracts, and how the company's products are gradually becoming requirements by regulatory bodies worldwide all decrease the downside risk of Genasys.

Genasys' software business represents a relatively minor percentage of total revenue at the present day. I am under the belief that the continued build-out of its software platform and the upcoming regulations on public safety will be catalysts leading to significant incremental value creation over the next few years and drive higher profitability, as seen by its Q3 2020 ~\$12mm in revenue (35% YoY growth) and ~54.1% gross margin. The latest closing price of Genasys was \$6.78. The share price has appreciated 100%+ from trading around \$3 at the beginning of 2020 and may no longer be undervalued, but this could be a growth play with more upside potential remaining. I believe the future growth (especially in the EU) and the value of the multi-year contracts are being ignored at the current share price.

Based on my valuation, I believe Genasys should be trading around ~\$8.15 (approximately 20% upside), and this positive upward trajectory should continue to gain momentum as we enter 2021 and get closer to 2022. By then, more countries, other than just the EU and US, would have increase regulatory requirements for PWS and LRAD related hardware and force adoption – thus, making this opportunity a promising long-term investment.

Sample Trading Initial Pitch

"The stock I would like to pitch is Medpace Holdings (NASDAQ: MEDP). They're a mid-sized contract research organization that performs clinical trials for biotech firms to get new drugs approved. Medpace primarily operates on a fee-for-service basis, meaning they get paid whether or not the drug is successful but get paid more for successful trials as those run longer and have more fees.

Medpace focuses on smaller biotech firms, which have seen a surge of venture capital funding. This VC funding pays for the services of Medpace to see if potential drugs can get FDA approval. There is a lag in timing, so the funding we are seeing now will show up on Medpace's future earnings. Compared to its peers, Medpace is well-positioned. Labor costs are a big input in Medpace's business but are lower when compared to its peers. Medpace has consistently delivered above averages, strong profit margins.

The stock is trading at \$116, which is in the middle of a wide 52-week trading range of \$58 to \$144. The current trailing or backward-looking P/E ratio is 41x, which feels high on a standalone basis, but is negatively impacted by COVID related lockdowns slowing existing clinical trials. On a normalized forward basis, it's a quite reasonable 24x. I believe that stronger revenue growth and EPS growth will drive the stock price higher. I think we can easily see the stock trading closer to \$130 in the next six months with annual EPS ticking up to \$4.33 with a 30x P/E ratio."

Notice how compared to the fundamental pitch, there's less of a focus on the company specific details and "story." This doesn't mean that the fundamentals of the business don't matter; instead, the takeaway is that most trading usually has a shorter-term outlook on the company. Traders, particularly those that specialize in complex financial derivatives or short-selling, don't have the luxury of being able to wait for a thesis to come true (usually a few months at most). Right now, Medpace is benefiting from a short-term, unexpected event, as seen in its share price growth since the beginning of 2020.

Let's say you're about to analyze an investment opportunity on a public company. What are some questions you would ask?

During your analysis, try to first understand where the company has been, where it's at the present day, and then attempt to build an investment thesis on where the company is headed.

- What is the company's mission statement and what value do they provide to their customers?
- What is the background of the management team, how long have they been running the company, and what is their track record?
- What is the business model? How does the company generate cash flow?
- What types of products/services does the company sell? Would you consider them to be valuable?
- What is the company's strategy to achieve growth? Has its strategy adapted alongside the market?
- What are the end markets to whom the company sells its products/services?
- How has the company's financial performance been in recent years?
- Is the company's cash flows sustainable? Does its revenue have a recurring component?
- What are the risks to the company's cash flows? How competitive is the market?
- What are the growth prospects of the cash flows? Where do you predict revenue growth and opportunities to create profitability to come from?
- What has earnings per share (EPS) been historically, and how has it been trending in recent years?

Question each of your assumptions carefully from the perspective that your thesis might be flawed, as doing so will help solidify your pitch as you understand the counter-arguments. By identifying the weaker areas of your pitch, you can defend your viewpoints better since you've looked at the investment from both perspectives with an open mind, as opposed to looking for what confirms your pre-existing beliefs.

What are some of the various investment strategies employed by hedge funds?

1. **Long-Short Equity:** Involving picking winners and losers based on the fundamental analysis of companies; The long-short offsetting pairs reduce the overall market risk.
2. **Event-Driven Investing:** Pursues pricing inefficiencies before or after major corporate events such as mergers, acquisitions, and spinoffs – either looking for potential M&A targets or seeking to trade the acquisition premium on announced acquisitions.
3. **Activist:** Activist funds seek underperforming companies they believe are poorly managed and attempt to be the catalyst for a successful turnaround.
4. **Quants:** Includes systematic, rules-based, algorithmic trading, proprietary strategies, momentum, mean reversion, and high-frequency trading (HFT).
5. **Arbitrage/Relative Value:** Seeks pricing inefficiencies/mispricing and trading spreads within an asset class (e.g., volatility arbitrage in the options market, convertible arbitrage in stock vs. convert trading).
6. **Long-Only:** Often associated with value investing, long-only funds only take long positions on the common equity of companies they believe are undervalued with a longer-term holding horizon.
7. **Short-Only:** Short-selling specialists pursue overvalued stocks. However, the most prominent short-only funds are associated with uncovering accounting frauds or other types of malfeasance.
8. **Market Neutral:** Like a long-short fund, but will pair long and short trades to mitigate market risk.

What is the investment rationale behind the long-short strategy?

When an investment firm goes “long,” it purchases a stock that it believes will increase to turn a profit. The opposite being the firm goes “short” on companies with share prices they expect to decrease. So if the share price goes down, the firm profits. Each equity investment contains some components of market and industry and company-specific risks. The objective of long-short investing is to hedge the market risk (i.e., cancel out the market risk) and then the industry and company-specific risk to some extent, so they're not on the wrong side in case the economy's trajectory suddenly flips.

How does a long-short fund differ from an equity market-neutral (EMN) fund?

A long-short fund and an equity market-neutral fund (“EMN”) are closely aligned, and at an initial glance, they appear to share the same goal. The difference is that a market-neutral fund will ensure the total value of long and short positions are close to being equal as possible to lower the portfolio risk, which is closer to the original intent of a hedge fund.

For this reason, market-neutral funds exhibit the lowest correlation with the market, and the goal of an equity market neutral fund is to generate returns independent of market movements. A distinct feature of a market-neutral fund is that it always yields positive returns, even if it could mean losing out on the more potential upside (i.e., limited upside). From a returns perspective, an EMN fund appears the least exciting overall, but it remains consistent with its objective of having no correlation to the equity markets.

Long-short hedge funds are similar in that they take long-short positions to hedge their investments, but they're more lenient on the portfolio's rebalancing. This means they don't require a perfect balance between longs and shorts, and if a certain market trend is performing well, they'll try to let the profits continue (even if the risk increases). Unlike EMN funds that have portfolio betas close to zero, long-short funds will have positive betas and have a direction while hedged (e.g., net long or net short) based on their market outlook.

Broadly, what is the purpose of hedging?

Hedging is a strategy used by an investor or a company to mitigate the risks of an investment. Hedging usually involves investing in derivative products that will be profitable if the market moves in the opposite direction the investor expects. While it may lower the potential upside, it provides downside protection in return.

What is the difference between gross and net exposure?

Two terms that are frequently thrown around in the hedge fund industry when discussing measures of risk are gross and net exposure. Gross exposure is a leverage metric, whereas net exposure is a market risk metric.

- **Gross Exposure:** Gross exposure is defined as the percentage of a portfolio invested in longs, plus the percentage that's short. If the gross exposure is in excess of 100%, the portfolio is levered (e.g., using borrowed funds).

$$\text{Gross Exposure} = \text{Long Exposure \%} + \text{Short Exposure \%}$$

- **Net Exposure:** Net exposure is calculated as the percentage of a portfolio invested in longs minus the percentage of the portfolio that's short.

$$\text{Net Exposure} = \text{Long Exposure \%} - \text{Short Exposure \%}$$

What is a maximum drawdown?

The maximum drawdown (MDD) is a measure of a portfolio's maximum downside risk over a period. The MDD is viewed as the maximum potential loss as the question being answered is: "What is the most an investor might lose?"

$$\text{MDD} = \frac{(\text{Peak Portfolio Value} - \text{Lowest Portfolio Value})}{\text{Lowest Portfolio Value}}$$

As shown above, the MDD formula shows the largest percentage drop in a portfolio's value based on the peak and lowest points of a portfolio. But be mindful that for MDD to be a meaningful measure of risk, the portfolio should have become active in the market for more than a decade at a minimum to ensure the portfolio has undergone at least one full economic cycle and experienced at least one major bear market.

How do investment strategies based on beta, enhanced beta, and alpha differ?

- **Beta:** Roughly, beta is the correlation of your portfolio with the broad market. If your portfolio has a beta of 1.0, then if the stock market goes up 10%, you would expect your portfolio to go up 10% too. Beta strategies track an underlying index, most often the S&P 500.
- **Enhanced Beta:** Strategies that track an underlying index but outperform Beta by adding an Alpha strategy (called "overlay"). Enhanced beta is still classified as passive investing and tracks an index, for the most part. However, there are some additional adjustments to the holdings to achieve greater returns, such as taking market capitalization and dividend payments into considerations (i.e., added element of active management to adjust the proportion of the weighting of specific stocks to outperform beta while remaining risk-averse).
- **Alpha:** In contrast, alpha is the excess return amount above the market return and measures the out-performance of a certain security or portfolio, achieved with above-market returns. Said another way, alpha involves the uncorrelated returns to the market. When portfolio managers discuss alpha, they're almost always referring to "beating the market."

Explain the merger arbitrage investment strategy?

The merger arbitrage investment strategy is when an investor purchases the shares of a company undergoing an M&A deal. The investor can take advantage of inefficiencies in pricing surrounding M&A, as it often takes time for the market to digest the recent information and settle on a valuation. The investor can profit from the uncertainty that exists as the market evaluates an announced acquisition. The overarching aim is to profit from the spread between the current market price and the announced acquisition price.

Read More → [Merger Arbitrage Example](#)

What is the event-driven turnaround investment strategy?

The goal is to capitalize on a particular event, ranging from a regulatory change to a full operational turnaround. The ideal investment characteristics are underperformance vs. peers with a strong, aligned management team. The right team is of high importance because they execute the turnaround strategy. The investor would examine the management team's track record and see if they have any relevant history in turnaround situations and what makes them the right leaders for the job.

What is activist investing?

In activist investing, poorly run and underperforming companies are targeted by turnaround specialists that will push for changes in either management practices or completely replace the management team. A key distinction is that the catalyst to the turnaround is the entrance of the activist investors themselves. The public announcement of an investment by a well-known activist investor alone can cause a company's share price to spike, as investors expect drastic changes to come.

For example, an activist investor could take a large equity stake in a company intending to drive the management's strategic decisions (or maybe even replace the team). Their goal is to convince existing shareholders that there's hidden value in the company that the current management is not extracting (and thus, the company is undervalued). Even though their investment is a minority stake, these activist investors can often change a company's trajectory (e.g., Starboard Value, Elliott Management) and influence an underperforming company.

What does convertible bond arbitrage mean?

The convertible bond arbitrage strategy is when an investor will take a long position on a convertible bond and then short the underlying stock. The investor would then benefit from the difference in pricing between the convertible bond and the share price.

If the share price declines, the investor can benefit from the short position taken, and thus there'll be more downside protection. But if the share price increases, the investor can convert the bond into shares and then sell, earning enough to cover the short position (and again minimize the downside).

What is momentum investing, and how does it differ from GARP?

Richard Driehaus, of Driehaus Capital Management, is often credited as being the father of momentum investing. The general theme of momentum investing is: "buy high, sell higher."

The momentum investing strategy strives to capitalize on the continuance of ongoing trends in the market – hoping the trend persists. The difference from GARP investing is that momentum investing relies more on technical indicators to decide trades, rather than being value-oriented and looking for high-growth but still undervalued companies. The holding period is shorter in momentum investing as it intends to take advantage of trends that are often driven by human emotion, which coincides with overvaluations of certain equities.

How does momentum investing differ from mean reversion?

- **Momentum Strategies:** Momentum-based strategies rely on the continuance of existing market trends to ride the upward trend. Recently, securities (or a specific asset class) have performed well, and these investors believe there's still more upside remaining for profits to be made.
- **Mean Reversion Strategies:** Mean reversion based strategies pursue unusually large drops or increases in pricing due to market over-reactions (either oversold or overbought) and then place bets that the price will revert towards the mean.

How does quant vs. human investment strategies compare?

Systematic Strategies

- **Rules-Based:** Investments are unaffected by emotion – the most frequently cited reason for misjudgment.
- **Data Capacity:** Systems can interpret and analyze more data and inputs in volume and speed.
- **Machine Learning (ML):** Systematic strategies will adapt and continuously improve over time.
- **Back Tested:** These systems can identify which specific strategies would have worked in the past to help guide future decisions (i.e., run simulations of how a strategy would have performed).

Human Traders

- **Adaptability:** Traders are more experienced, can adjust to market changes, and can pick up sudden sentiment changes that computers cannot capture.
- **More Resourceful:** Capable of capturing more market color and feedback that's not in electronic format (e.g., salesperson calls, investor presentation meetings).

How does a hedge fund with a global macro strategy invest?

A hedge fund with a global macro strategy bases its investments on the overall economic and political environment. The relations between countries and macroeconomic trends can affect numerous financial instruments. Thus, a global macro fund's holdings are more diverse relative to traditional hedge funds and include common equity, fixed income, currencies, futures on commodities, and other specialized options.

The strategy will shift based on changes in economic policies, global events, and changes in foreign policies/regulations as these present the investment opportunities that these global macro funds seek and then actively pursue. The fund's focus will often be on liquid broad indices rather than specific equities, which is why macro hedge funds are very active in rates, currencies/FX, commodities, and equity indices.

Most large institutional asset management firms are multi-strat. What does this mean?

Most institutional investors that manage billions of dollars, such as Blackrock, are considered multi-strat, which is a catch-all term for having a diversified combination of investment strategies. Given the various strategies employed, this type of approach provides more stability in returns and has more downside protection. For example, a multi-strat investment firm will invest in equities, bonds, real estate, and private equity to spread its capital allocation and de-risk its portfolio holdings. Considering their AUM, capital preservation often takes priority over achieving outsized returns.

Why would it be incorrect to compare a hedge fund's returns to the market return?

Traditionally, hedge funds were never intended to be investment vehicles to "outperform" the markets. Instead, hedge funds provided their investors with decent risk-adjusted returns regardless of the market conditions. As the name suggests, they serve as a "hedge" against the market, and wealthy investors would park a small portion of their wealth into funds, rather than placing 90%+ of their net worth into these funds, as many people falsely assume that they do.

During bull markets, funds would have relatively decent returns, whereas, during bear markets, funds would again have decent returns. Many investors, particularly those in favor of long-term value investing, criticize the high active management fees and argue that index funds would be a better strategy, but under the false presumption that hedge fund LPs share the same belief system as them.

However, over the years, the objective of hedge funds has significantly shifted, and many fund managers explicitly aim to outperform the market in both bull and bear markets.

Why does it become more difficult to achieve high returns as a fund grows in AUM?

If the assets under management (AUM) of a firm grow, achieving high returns becomes increasingly difficult. The reason being large firms are considered “market movers,” meaning each of their actions is closely followed and the sheer size of their investments alone can cause the pricing of a stock to move up or down. For example, if a large firm sells its shares, other investors in the market could assume they're selling for a reason (i.e., under the belief the firm has more information than them), purchase demand will shrink, and the price will fall. Given the reasons above, large asset managers cannot invest in small-cap stocks, and instead, they're limited to invest in large-cap stocks since the impact their decisions have is lessened. But because large-cap stocks are widely followed by equity research analysts and investors alike, these assets are more efficiently priced.

Hence, many small-cap oriented funds place a cap on their AUM, even if they could raise more capital. These firms specialize in small-cap companies, where they believe the likelihood of finding mispriced securities is higher and they could achieve the highest returns for their investors.

Tell me about the Modern Portfolio Theory (MPT).

Introduced by Economist Harry Markowitz, the premise on which the Modern Portfolio Theory (MPT) is built upon is that “efficient portfolios” can be constructed in which the expected return can be maximized for a level of market risk with the proper allocation. Simply put, MPT is maximizing the return investors could achieve from their portfolio while bearing the least amount of risk possible (for their targeted return).

A central idea of MPT is that an asset shouldn't be assessed as a standalone investment from a risk standpoint but based on its contribution to the overall portfolio. Meaning, the portfolio beta matters more when selecting investments for a portfolio, rather than considering only the asset's individual beta. The relationship between risk and return of various portfolios would be plotted to determine the optimal asset allocation at each risk level where the expected return is maximized. Although systematic risk is unavoidable, diversified assets in a portfolio that are not correlated with one another can reduce the unsystematic risk of a portfolio. In a well-diversified portfolio, the total risk in the portfolio should be mostly systematic risk.

What is the Sharpe ratio and why does it have such popularity among investors?

The Sharpe ratio was developed by William Sharpe, who is also responsible for the CAPM model. The ratio is one of the most widely referenced metrics in academics, certain institutional investors, and everyday investors due to its simplicity. The use case is to assess the risk/return of a portfolio, or an individual investment, although most use it for a group of investments (e.g., mutual funds).

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

The Sharpe ratio begins by subtracting the risk-free rate to isolate the excess return of the portfolio. The excess return is then divided by the portfolio's standard deviation (a proxy for portfolio risk). A ratio below 1.0 would be considered a sub-par portfolio return considering the risk undertaken, while a ratio above 1.0 being acceptable and beyond 2.0 is rated as a well-performing portfolio on a risk-adjusted return basis (with 3.0+ being exceptional).

The frequent critique of the Sharpe ratio is how total portfolio risk is proxied by the portfolio's standard deviation, which is an oversimplified assumption that all equity returns are on a normal distribution. For this reason, there are many variations of the Sharpe ratio that account for portfolio risk in different ways, such as the Sortino ratio, which uses downside deviation rather than the total standard deviation of portfolio returns.

Why do many investors disregard equity research reports?

An equity research report is a document that provides a formal recommendation on whether investors should buy, hold, or sell shares in a specific public company. However, most equity research analysts have an inherent bias to be more positive about a stock. Equity research analysts have an incentive to give “buy” recommendations to maintain close relationships with the company's management teams being assessed. If an equity research analyst scrutinizes a company, it may not take future calls from that analyst. The investment banking division may also indirectly benefit if the company has a close relationship with the equity research division. This could lead to assignments helping the company if it raises capital or makes an acquisition in the future. The skewed rating system is well known across the market; for example, if an analyst issues a “hold” recommendation for a company, this is usually interpreted as a very negative signal by the investor community. In addition, equity research analysts are legally restricted from investing in the companies under their coverage group due to regulations to prevent a conflict of interest. For these reasons, it's important for investors to not take equity research reports at face value and instead make their own decisions.

What would happen if the US stopped requiring quarterly earnings reports?

Under SEC requirements, public companies in the US must file quarterly reports. This became a topic of debate after many prominent CEOs and investors such as Jamie Dimon, Elon Musk, and Warren Buffett expressed their agreement with modifying the quarterly filing requirement, under the reasoning that quarterly reporting places too much pressure on the management teams to meet quarterly EPS and earnings expectations.

Counter-Arguments

- Management teams that desire longer gaps between reporting periods could be hoping to conceal their poor performance. Although many acknowledge quarterly reports can lead to short-term oriented decision-making at the expense of a long-term strategy, one can attest that hitting short-term targets is proof that a long-term strategy is working. Any company can make poor choices, underperform, and then claim they're employing a long-term strategy. If short-term EPS targets distract a qualified management team from wise decision-making, they likely are not qualified to be running a public company.
- Shareholders are owners of the business and would create further distance between managers and shareholders (i.e., worsen the “agency problem”). As partial owners, shareholders should have the right to be aware of recent performance and company news.
- The lack of available information on what's going on within companies could make equities less attractive to risk-averse investors. The market would become less efficient since less public information is available. Thereby, market volatility would decrease during non-reporting periods while reporting periods would see substantial price movements (i.e., leading to instability in markets).

Walk me through the process of a hedge fund shorting a stock.

Short selling is when an investor predicts a short-term decline in the price of a particular security. For an investor to short a stock, it means the investor has “borrowed” the security from a broker and sold it in the open market, under the belief they'll repurchase the shares for a lower price later.

1. First, the hedge funds will go onto the exchange and sell the shares that they have borrowed from a securities lender or broker – meaning, they sold a stock that they don't actually own on paper.
2. Once the hedge fund wants to exit their short position, they'll re-purchase the shares and return the shares they bought to the lender.
3. If the short went as planned (i.e., the share price dropped), then the hedge fund's profit will be the difference between the purchase and sale price, less any associated fees paid to the lender – so the hedge fund has bought it back at a lower price than what they had sold the shares.

Name two of the key metrics to assess short interest.

1. **Short Interest Ratio:** Total number of shares shorted and open positions not bought back as reported by the exchange, divided by average trading volume

$$\text{Short Interest Ratio} = \frac{\text{Short Interest}}{\text{Average Trading Volume}}$$

2. **Short % of Free Float:** Short interest as a percentage of Free float, or the number of shares available to be traded (excludes insider owned shares)

$$\text{Short \% of Free Float} = \frac{\text{Short Interest}}{\text{Free Float}}$$

Tell me about the difference between index shorting and alpha shorting.

- **Index Shorting:** Index shorting can involve shorting an index such as the S&P 500 or buying put options on an index to hedge out the long positions.
- **Alpha Shorting:** Alpha shorting refers to taking individual short positions to make a profit. Funds will employ both approaches, but alpha shorting is much more difficult.

What is the difference between shorting a stock and purchasing a put option?

For shorting a stock, the investor has unlimited downside potential. Recall that short-selling means the investor sold a stock that they technically don't actually own, believing that they'll purchase it for a lower price in the future (i.e., the potential share price upside is unlimited). When purchasing put options, the maximum loss is limited to the initial investment amount.

Could you explain what a short squeeze is?

A short squeeze occurs when there's much short interest on a particular stock, and then positive news comes out. Many short sellers simultaneously attempt to cover their short positions at once, which causes a rapid increase in the share price. A short squeeze accelerates a share price's upward movement as short-sellers attempt to buy back the stock and cut their losses.

List some characteristics of an ideal short.

- Complacent incumbents facing a threat from innovative new entrants – must assess the management's willingness to adapt and the likelihood of a proper reaction (e.g., Nokia's failure to adjust).
- Companies focused on consumer euphoria – significant short-term interest in a consumer-oriented product (i.e., the demand is driven by a consumer-led "fad").
- Incumbents that are unlikely to exist in the future (e.g., physical newspaper business) and don't appear to be making any effort to adjust to the changing environment.
- Signs of accounting fraud or allegations made by credible sources

Why do many investors pay such close attention to special situations?

Special situations refer to specific event catalysts that could cause significant price movement in the near-term (or long-term) future. The term special situation is an umbrella term for many transactional, corporate restructuring, or financial engineering efforts completed by companies. The common theme among these examples is they present potential arbitrage opportunities. There is significant empirical evidence suggesting that spin-offs, split-offs, and divestitures can present positive value investment opportunities.

Many examples of special situations are related to M&A, including spin-offs, split-offs, merger arbitrage, and tender offers. Others are related to internal decisions such as new management teams, restructuring plans (turnaround strategy), recapitalizations, and more.

Specifically, what does it mean when an investor is a “contrarian” investor?

The simplest answer is that contrarian investors attempt to find investment opportunities that go against the consensus view. But the more detailed response is that contrarian investors invest in companies the market has given up on, either following poor performance or because future prospects appear bleak. The contrarian investor is implicitly investing under the belief that the markets over-react or neglect certain factors that have distorted a company's pricing.

A contrarian investor nearly always takes on a contrarian mindset when searching for investments, but the common misconception is that they constantly invest differently just for the sake of being a contrarian. Instead, contrarians are extremely patient and rarely make an investment unless they have sufficient conviction to take on this bet against the consensus. The basic principle of being a contrarian investor is the ability to think for oneself and being able to develop a set of views independent of what other people around you believe or want you to think. While this quote was said in the context of VC investing, Reid Hoffman, the co-founder of LinkedIn and partner at Greylock, said, “It's actually pretty easy to be a contrarian. It's hard to be contrarian and right.”

Why have so many hedge funds closed in recent years?

In aggregate, hedge fund returns have compressed in recent times and underperformed the S&P 500 over the past decade. The increased AUM in the market has made it more difficult to find good investment opportunities. Because of this, many large, brand-name funds have closed and returned capital to their investors. And given the increased competition, some ways that hedge funds are becoming increasingly sophisticated are the usage of satellite imagery, point-of-sale data, credit card data, and the employment of data scientists. The need for a hedge fund to be differentiated and carve out a unique niche to achieve outsized returns has never been greater – but this is far easier said than done in reality.

What is the purpose of diversification in portfolio management?

Portfolio diversification is when a portfolio contains investments in diverse industries, asset classes, and geographies. The end objective is to limit exposure to one asset class or particular risk and construct a portfolio of holdings that are not correlated while still attempting to maximize returns (i.e., nearly all unsystematic risk is eliminated). The chosen investments shouldn't be correlated with one another, or else that would defeat the purpose of diversification.

Broadly, could you define what value investing entails?

Value investing could be best described as the strategy of finding temporarily mispriced stocks trading for less than their intrinsic values, under the belief that the market is undervaluing these companies. From their viewpoint, these opportunities arise because the market is emotional and often overreacts to good or bad news, resulting in share price movements that incorrectly align with the company's long-term fundamentals in question. By purchasing equities at deflated prices, they seek to profit from share price appreciation once the market corrects itself.

To recap, value investing is essentially attempting to capture some level of disconnect between the current trading price and the company's true intrinsic value (that's not immediately apparent to the market). The value investor will then attempt to identify the catalyst that would make the market realize the company is undervalued, but by then, the investor would have purchased the security at the discounted price.

What does it mean when a security has a sufficient margin of safety?

A margin of safety is one of the fundamental concepts of value investing in which a security is purchased only when the share price is significantly below its intrinsic value. The margin of safety serves as an investor's downside risk protection and can be thought of as the difference between the estimated intrinsic value and the current share price.

Said another way, the margin of safety is the buffer that value investors place into their investment decision making to protect them against downside risk if the share price drops post-investment. Rather than shorting stocks or purchasing puts, many value investors view this as their primary method to mitigate investment risk.

$$\text{Margin of Safety (MOS)} = 1 - \left(\frac{\text{Current Share Price}}{\text{Intrinsic Value}} \right)$$

Often, many value investors will not invest in a security unless the margin of safety is ~20-30%. If the MOS hurdle is 20%, the investor will only purchase a security if the current share price is 20% below the intrinsic value based on their valuation.

How do you determine whether a company has an “economic moat” or not?

An “economic moat” is the competitive advantage of a company and the factors that protect its profits from competitors. The moat will be evident in the company’s unit economics (e.g., consistent operational performance, high margins) and is caused by any unique advantages. Such as patents, proprietary technology, branding/reputation, product or service value, switching costs, or network effects. These traits shouldn't be easy to replicate by other competitors in the market and come with barriers such as high switching costs or capital requirements.

Said another way, an economic moat refers to a company’s ability to maintain its competitive edge over its competitors to protect its continued long-term profit generation. The stronger and more defensible a company’s competitive advantage, the more difficult it becomes for other new entrants to breach this moat and take away their market share. An economic moat can be viewed as protecting the competitive positioning of a business.

If a company has an “economic moat,” sustainable value creation over the long-term is attainable.

Read More → *Measuring the Moat: Assessing the Magnitude and Sustainability of Value Creation*

Why would an investor use the “Scuttlebutt method”?

Coined by Phil Fisher in his book “Common Stocks and Uncommon Profits,” the Scuttlebutt method refers to investing based around going out and conducting your own proprietary research (i.e., listen for rumors, observe surrounding developments, observe stores), rather than just reading equity research reports and financial news, and relying on the opinions of others. This type of diligence involves talking to all kinds of people (e.g., customers, employees, industry experts) to gain their insights and identify trends by gathering your information and then making investment decisions based on your intuition.

Why does Warren Buffett dislike the EBITDA metric?

Warren Buffett and Charlie Munger are both known for having a particular distaste for management teams and equity research analysts using the EBITDA metric. In Berkshire Hathaway’s 2000 shareholder letter, Buffett famously wrote, “References to EBITDA make us shudder – does management think the tooth fairy pays for capital expenditures? We’re very suspicious of any accounting methodology that’s vague or unclear since too often that means management wishes to hide something.” Given how EBITDA neglects capex, Buffett doesn’t believe EBITDA is a true representation of a company’s financial performance. Failing to account for capex would particularly apply to capital-intensive industries such as manufacturing.

What is Warren Buffett’s view on the capital asset pricing model (CAPM)?

Warren Buffett has a negative view of the capital asset pricing model (CAPM) due to his disagreement with beta as a measure of risk. From his perspective, CAPM is ineffective because it mistakes volatility as a risk when in reality, it could represent an opportunity to purchase undervalued securities (rather than being a sign it’s a risky, less valuable investment that should be avoided). Buffett has been very outspoken about this topic, going

as far as calling beta nonsense, and has stated how in his and Charlie Munger's opinion, "Volatility is no measure of risk to us." Instead, Buffett defined risk as "the possibility of harm or injury."

Contrary to what's taught in academic literature, volatility doesn't equate to risk according to Buffett. This means that the higher the chance you might lose your initial capital, the riskier the investment is. How you determine the probability of losing your capital is based on the quality of the fundamentals of the business, the margin of safety (pricing vs. intrinsic value), and doing enough research into every aspect of the business as risk comes from ignorance.

What is the difference between value investing and deep value investing?

Value investing and deep value investing are price-focused strategies to profit from companies trading at a discount to their NPV. While the definition is subjective, deep value investors are distinct for refusing to pay a higher price for quality. This means they're often willing to make reasonable bets on cheap equities despite a clear deterioration in quality, including weak recent financials, ineffective growth strategies, and incompetent management teams. The reasoning behind this is that deep value investors believe quality is already priced in.

Thus, the only way to make outsized returns in the market is by searching for opportunities where the investors have over-reacted to bad news (often terrible news) on underperforming companies. Many deep value investors speculate on a new turnaround strategy once rock bottom has been reached, as this would imply a restructuring of some sort, such as a new management team and the sale of non-core assets. An activist investor may frequently step in and help initiate a change of course for the business, precisely the type of catalyst the deep value investor is waiting on.

How do value investors deal with volatility in share price movement?

Simply put, value investors have a long-term thesis and therefore ignore short-term movements. Often, a value investor will hold a security for 5+ years and invest with the mindset of "never selling."

Explain to me what the purpose of dollar-cost averaging (DCA) is.

Commonly recommended by practitioners of value investing, the practice of dollar-cost averaging (DCA) refers to committing to buying shares at pre-determined intervals. This can be useful during periods of market volatility and prevent an investor from attempting to "time the bottom."

If the stock price decreases, then the investor can simply purchase more to lower the average cost paid on a per-share basis. But if the price increases, the investor can either continue purchasing (if still undervalued based on their beliefs) or pursue other opportunities in the market.

Explain the growth at a reasonable price (GARP) investment strategy?

Growth at a reasonable price (GARP) is a hybrid investment strategy that combines growth investing and value investing. GARP could be best described as profiting from a company trading at a material discount on an earnings basis but not to the same extent as traditional value investments.

The GARP investment strategy is based on finding high-quality companies with a sustainable competitive advantage, above-market consistent earnings growth, and large market potential. Then, the NPV of the cash flows will be determined with the growth potential incorporated to see if the company is undervalued. While this is not necessarily pure value investing, it's a reasonable compromise that many investors have adopted.

What are value traps?

Investors new to value investing will frequently fall into the so-called value trap. This fallacy is the false belief that a particular stock is currently undervalued because it has fallen out of favor with the market and dropped significantly in recent weeks or months. Because these investments appear to trade at such low levels

compared to the prior levels, many investors will view this as a buying opportunity. However, the fall in price is misleading, as there's usually a good explanation for the decrease in valuation.

Explain the efficient market hypothesis (EMH) and its three forms?

Introduced by Eugene Fama, the efficient market hypothesis (EMH) theory asserts that asset prices fully reflect all available information. Following the release of new information or relevant data, prices will automatically adjust instantaneously to reflect the accurate price within a matter of seconds.

- **Weak Form EMH:** The weak form of the EMH states that all past information such as historical trading prices and volume data is reflected in current asset prices.
- **Semi-Strong EMH:** The semi-strong form of the EMH states that all publicly available information is reflected in current asset prices.
- **Strong Form EMH:** The strong form of the EMH states that all public and private information (including inside information) is reflected in current asset prices.

If the efficient market hypothesis were true, what implication would this have on active management?

Since the EMH theory contends the current prices reflected in the market already reflect all information, an attempt by an investment manager to outperform the market by finding mispriced securities or timing the performance of a certain asset class in an industry is a game of chance rather than skill. If the efficient market hypothesis were strong form efficient, there would be no point in active management.

It is important to note that EMH by its definition is referring to the long-run. Therefore, if an investor purchased a temporarily undervalued stock (by their definition) and then sold for an above-market return – that doesn't invalidate the EMH theory. Rather, most proponents of EMH suggest that these types of occurrences are rare and not worth the effort (and active management fees) over the long-term.

Why has passive investing become more popular for everyday investors?

Index funds such as mutual funds and exchange-traded funds (ETFs) are convenient for the everyday investor to participate in the markets. The widely held belief among the passive investing community is that it's very difficult to beat the market, and it would be a futile attempt when investment professionals that spend considerable resources and time on analyzing securities struggle to do so consistently.

How does the random walk theory differ from the efficient market hypothesis?

The random walk theory similarly concludes that attempts to predict share price movements accurately are futile, and past successes are due to luck rather than actual skill. The random walk theory is a hypothesis that assumes the share price changes in the financial markets are due to random, unpredictable events that nobody can predict. In contrast, EMH is the theory that asset prices perfectly reflect all the information available in the market. Under this assumption, a company's share price can neither be undervalued nor overvalued, but it's precisely trading where it should be because the market is efficient.

Sales & Trading

What is the role of the sales & trading division in an investment bank?

Sales & Trading is the division of an investment bank responsible for selling issuances of stocks, bonds, and derivatives.

- **Sales Division:** The salespeople work more closely with institutional investors such as large asset managers, hedge funds, insurance companies, pension funds, and other buy-side investors to pitch ideas and buy or sell securities or derivatives. Simply put, the sales division manages the communication and relationship building with clients on behalf of the investment bank.
- **Trading Division:** Traders make a market and execute trades on behalf of the investors they represent. Unique to this role, sometimes traders have their own trading book where they can take positions and generate a personal P&L. Traders need to be quick with mental math, have the quantitative skills to understand complex products, and have an intuitive understanding of markets to spot mispricing.

What is a derivative?

A derivative refers to a financial instrument in which its value is derived from an underlying asset or index. Some of the most common derivatives include options, futures contracts, forwards, and swaps. For example, a call option tied to a particular stock would be an example of a derivative, since the option's value depends on the company's share price.

What are options?

Options are financial derivative instruments that give the holder the right, but not the obligation, to purchase or sell an underlying asset at an agreed-upon price. This right will have a set expiration date upon which, if the option is not exercised, it'll expire worthless.

What is the difference between a call and a put option?

- **Call Options:** Call options are financial contracts that give the owner the right, but not the obligation, to buy an underlying asset at a pre-specified price within a period. If the option holder believes the price of the underlying asset will increase ("bullish"), they'll purchase calls because the value of a call option will increase as the underlying asset's value increases.
- **Put Options:** Put options, in contrast, are contracts that give the owner the right, but not the obligation, to sell an underlying asset at a pre-specified price within a period. If the option holder believes the price of the underlying asset will decrease ("bearish"), they'll purchase puts since the value of a put option increases as the underlying asset's value decreases.

Tell me the differences among options that are "in-the-money," "out-of-the-money," and "at-the-money."

- **In-the-Money ("ITM"):** If an option is "in-the-money" (ITM), this means that the option is profitable to exercise. For a call option, an ITM option would mean that the option price of the underlying asset is above the exercise price (or strike price).
- **Out-of-the-Money ("OTM"):** If an option cannot be exercised profitably, it's referred to as being "out-of-the-money" since the option has no intrinsic value.
- **At-the-Money ("ATM"):** When an option is at-the-money, this means the option's strike price is equal to the price of the underlying security. An ATM option, similar to an OTM option, has no intrinsic value but could still have time value depending on the expiration date.

What are cash equities trading groups at investment banks?

Within investment banks, the equities groups are split between cash equities and derivatives. The term “cash equities” is another term for stock trading commonly used when discussing sales & trading. The set-up of a cash equities group is completely different today due to the diminished requirements for actual traders on the floor or market makers due to the reliance on electronic trading. Nowadays, most investment banks have only a handful of cash equity traders globally, as opposed to the hundreds formally employed in the last decade.

What does a book of orders consist of on trading exchanges?

During the period of open-trading (or “continuous market-making”), there's a book of orders that contains bids and offers.

- **Bids:** Bids are when there's interest to purchase the security from buyers.
- **Offers:** Offers are when existing holders are interested in selling the security to the open market.

Each order has a price and the number of shares at that specific price. Orders are then automatically ranked, with the highest bid at the top and the lowest sell offer at the top.

What is the indicative cross size?

The indicative cross size calculates the number of shares that can connect buyers with sellers at a specific price point. The closing price would be the price with the highest cross size (i.e., the price at which you can get the maximum number of buyers and sellers to exchange shares).

List some key volume indicators used by traders.

- **Volume:** The total number of shares that are traded within a day. A stock with a higher average daily volume means you can trade more shares before moving the price.
- **VWAP:** VWAP stands for the “volume-weighted average price.” This indicator measures the average price based on actual trading volumes.
- **Depth:** The depth reflects the total buy and sell orders currently on the order book and is used to measure how much one could purchase in terms of volume without “moving markets.”

How does an order get routed?

There are multiple venues for a sales trader to send an order to:

- **Order to the Floor:** The exchange (NYSE or NASDAQ) where the stock is listed.
- **Internalization:** The bank executing the trade could provide the other side, offsetting the trade flow with their trade flows.
- **Electronic Communication Network:** ECN is a computerized stock exchange that connects buyers with sellers. Stocks are not listed on an ECN, but the ECN pays rebates taken out of the bid-offer spread to participants, which provides liquidity to the ECN.
- **Smart Order Routing (SOR):** Alternatively, SOR algorithms can scan the market and spread the order through multiple trading venues.

What are some examples of off-exchange trading platforms?

- **Block Trades:** A large number of shares are traded off the exchange, reporting requirements to FINRA (ticker, size, and the price paid), and typically traded after the market close.
- **Dark Pools:** Platforms to allow large trades to be executed without moving prices. Has reporting requirements similar to block trades, lower execution fees than exchanges, and flexibility in order types.
- **OTCBB/Pink Sheets:** Over-the-Counter Market (not exchange-traded) can be used by smaller companies that don't meet the exchange listing requirements but still need to file with the SEC.

What is a stock index and how can one be traded in the public markets?

A stock index is just a tracker of a collection of equities, reflecting the price of stocks included in the index and calculated based on the index provider's rules. The index by itself is not tradable on an exchange, although there are financial instruments such as ETFs that follow these major indices (e.g., SPY, VTI, QQQ).

What are the key equity indices and their unique traits?

S&P 500 (Bloomberg Ticker: SPX)

- 500 Large Cap US Stocks selected by Standard and Poor's (S&P)
- Minimum Market Capitalization of \$8.2 Billion
- Price Return Index (Total Return Index Available)
- Market Capitalization Weighted (i.e., Float Adjusted)
- Updated Quarterly to Add or Remove Constituents (and Update Share Counts for Market Cap Weighting)

DJIA (Bloomberg Ticker: INDU)

- Stands for "Dow Jones Industrial Average"
- 30 Large-Cap US Stocks (excluding Transportation and Utilities) chosen by S&P and the Wall Street Journal
- Price Return Index: Price-Weighted
- During stock splits, the share prices are multiplied by a divisor to reflect the split (divisor is ~0.147)

NASDAQ 100 (Bloomberg Ticker: NDX)

- Largest 100 Stocks Listed on the NASDAQ
- Ranked in Order of Market Cap Size
- Price Return Index (Total Return Index Available)
- Market Capitalization Weighted (Not Float Adjusted)
- Maximum Weights Rebalanced Quarterly

What is a mutual fund?

A mutual fund is an investment vehicle for both retail and institutional investors, in which investors are pooled together and each own shares of the fund proportional to the amount invested. Most mutual funds are open-ended, allowing for investors to increase or decrease their investment.

Mutual funds are typically bought and sold at the end of the day, at the fund's net asset value (NAV). The NAV is the total value of all the securities in the fund, plus cash, divided by the number of shares. It's the value of each mutual fund share as determined by closing market prices. A few traits to be aware of are that mutual funds are regulated by the SEC and are managed by an asset manager.

Could you define what an asset manager is?

An example of an asset manager would be one that actively manages a mutual fund on behalf of the fund's investors. These asset managers are considered fiduciaries, meaning that they're hired and compensated for advising their investors and making recommendations and/or investments in the best interests of their clients.

How do hedge funds differ from mutual funds?

The distinction between hedge funds and mutual funds is that hedge funds are not targeted towards retail investors. Instead, only accredited investors can invest as LPs in hedge funds. Hedge funds come with higher risks of losing capital. Thus, these strict qualification rules were put in place to protect the interests of those that can't afford to lose a significant sum of money. In addition, hedge funds are more actively managed, use different strategies, and are resultingly compensated more due to the "2 and 20" compensation model.

Would sales & traders be considered as fiduciaries?

No, sales and trading professionals are not considered being fiduciaries. Instead, sales and traders work directly with institutional and high-wealth investors that don't require (or request) advisory services. The primary role of S&T is to execute orders on behalf of clients without providing investment advice.

What is an ETF?

An exchange-traded fund (ETF) is a type of financial product that tracks a specific underlying index (i.e., collection of securities, most often equities). An ETF can be viewed as a blend between a mutual fund (portfolio of investments, passive strategy, benchmarked to index) while being similar to equities by being listed and traded on exchanges. ETFs have become a very popular choice recently due to lower fees and convenience.

What is the difference between the initial margin vs. the variation margin?

Initial Margin

- A fixed amount you need to pay to cover your credit risk, set by the Clearing Bank's Risk Department, which may allow offsetting positions
- Varies based on the volatility of the underlying product
- Covers the cure period risk of a clearing client not making their variation margin

Variation Margin (or Maintenance Margin)

- The Market-to-Market gain or loss you have on your position
- Cash or collateral is received if your position is in-the-money, but you pay cash or collateral if your position is out-of-the-money

What purpose do clearing houses serve in the derivatives market?

Following 2008, the Dodd-Frank Act required most derivatives (both exchange-traded and over-the-counter) to be formally cleared. Originally, the buyer or seller of an exchange-traded derivative would not be aware of who the counterparty is – but now, clearing houses are required to intermediate the transaction and help mitigate the credit risk. While the counterparty risk associated with derivative instruments is not eliminated by any means, the facilitation of the exchange by a 3rd party seeks to minimize it.

What is securitization?

Securitization is when financial instruments are pooled together into one group to become more marketable. The issuer will then sell this pool of repackaged assets to investors (e.g., mortgages pooled together and divided into securitized bonds). By themselves, the assets within the group would not have sufficient liquidity to be sold as a standalone product, but securitization can allow for the grouped assets to be easily tradeable in the market and thus have high liquidity.

What are collateralized debt obligations (CDOs)?

A collateralized debt obligation (CDO) is a structured-financial instrument backed by a collateral pool, which can include loans and other assets. These underlying assets are offered as collateral in case the loan defaults. Often put together by investment banks, the bankers will collect cash flow-generative assets (e.g., mortgages, bonds, debt) and then repackage them into a pool and customize it based on the level of credit risk the investor is comfortable taking on.

The most well-known example of a CDO would be mortgage-backed securities (MBS), a type of asset-backed security (ABS) secured by a collection of mortgages. MBS that consisted of subprime loans played a central role in the housing financial crisis and revealed one weakness of the securitization process: the lack of transparency in the assets pooled. Many felt the combination of collateral concealed the lack of quality to make the MBS

appear less risky. In addition, securitization allowed for mortgage originators to originate new, low-quality mortgages and then quickly offload them through an MBS offering.

Give me an overview of the mortgage-backed securities market.

Mortgage-backed securities are a very common way to finance mortgages in the US, UK, and Netherlands. The US MBS market is very large, even of greater size than the corporate bond market. Mortgage loans will first be placed into a special purpose vehicle (SPV), a holding company for the mortgages. The SPV will then issue MBS bonds. As the homeowners that took out the underlying loans make mortgage payments, they go to the MBS bondholders.

How is the MBS market split in the US?

1. **Agency Mortgages:** A government agency (e.g., Freddie Mac or Fannie Mae) guarantees the mortgage loan. If the underlying homeowner defaults, the agency takes on the credit risk and repays the loan.
2. **Non-Agency Mortgages:** There is no guarantee, and the bond investors take the credit risk of the underlying borrowers. This risk is reflected in the higher interest rate these borrowers pay, as well as over-collateralization, having a higher value of loans than MBS bonds issued.

What is an asset-backed security (ABS), and how do they differ from MBS?

Asset-backed securities (ABS) function similarly to mortgage-backed securities (MBS). Although mortgages are technically classified as an “asset,” the differences lie in how ABS is typically used to refer to financing (e.g., credit card loans, car loans or leases, student debt). In comparison, the ABS market is smaller than the MBS market since the borrowings related to ABS are typically smaller than mortgages, especially in the US.

What are collateralized loan obligations (CLO)?

Collateralized loan obligations (CLOs) are securities backed by a pool of low-rated corporate loans. CLOs are structured using low-credit rated corporate loans that are bundled together. The CLOs will have many tranches to appeal to different investors from a risk perspective. CLOs would be an example of a CDO but are unique in that usually just business loans are involved, and each distinct class of owners will receive differing yields based on the tranche and risk they're undertaking.

What is a forward contract?

Forward contracts are formal agreements between two parties in which one party agrees to purchase an underlying asset from the other party at a later date. The future date of purchase and the price at which the purchase will be made will both be stated in the original agreement.

What are futures contracts?

Futures are derivative instruments that serve as a contractual obligation for two parties to exchange an underlying asset at a pre-determined price at a later date. This exchange is completed regardless of the change in the asset's price (whether up or down). The buyer must either buy the underlying asset, or the seller must sell the underlying asset at the pre-specified set price.

Historically, futures were most common with commodities such as bushels of corn. Physical commodities would be physically settled, meaning the commodity (e.g., corn, barrels of oil, lumber) would actually be delivered in person. But in today's market, equity and interest rate futures have significantly higher trading volumes in comparison, which don't involve any physical delivery since they are monetary exchanges (i.e., cash-settled based on the value difference).

What is the difference between forwards and futures contracts?

The two derivatives, forward and futures contracts, are similar in that both involve agreements between two parties to buy or sell an underlying asset at a pre-determined price by a specified date. The difference is that a forward contract is a private agreement that settles at the end of the agreement term and is traded over-the-counter. A futures contract has more standardized terms (i.e., less customization) and is traded on exchanges, in which the changes in prices can be readily seen daily until the end of the contract term.

How does the spot exchange rate and forward exchange rate differ?

- **Spot Exchange Rate:** The spot exchange rate is the exchange ratio of one currency into another currency on a particular day. The key characteristics of spot exchange rates are that they're reported on a real-time basis and determined by the supply/demand of that currency relative to the same supply/demand dynamics of the other currency (i.e., set by the market).
- **Forward Exchange Rates:** In forward exchange rates, two parties will exchange currency at a specific date in the future. These deals are executed to hedge against foreign exchange risk, to which many multinational companies are exposed. Forward exchange rates can be quoted for limited durations (e.g., 30 days, 60 days, 90 days, 180 days).
- **Distinction:** The key distinction is that the rate is negotiated and agreed upon by the two parties for forward exchange rates (rather than letting the market determine the rate).

What is the difference between a warrant and a stock option?

- **Warrants:** A warrant gives the holder the right, but not the obligation, to purchase common shares of stock directly from the issuing company at a fixed price for a pre-determined period. The key distinction is that the warrant is issued directly by a company to an investor. Therefore, if the warrant is exercised, the shares required to fulfill the agreement are issued, so warrants have more of a dilutive impact when exercised because the company will actually issue new stock.
- **Stock Options:** A stock option gives the holder the right, but not the obligation, to purchase or sell shares at a pre-specified price for a defined period. Options are commonly traded amongst investors. The exercising of a stock option has less of a dilutive impact because no new shares are issued since options are derivatives based on an underlying asset (i.e., the pre-existing common shares of the company).

What is implied volatility?

The implied volatility ("IV") is the market's expectations on the movement in a specific security or indexes' price. Implied volatility has a significant role in the pricing of options contracts, as higher IV results in higher premiums (and vice versa). During periods of bearish sentiment, implied volatility typically increases, whereas when the market is bullish, the implied volatility decreases.

If the implied volatility were to rise, how would option prices be affected?

The implied volatility has a significant impact on the options market. The higher the volatility of the underlying asset (e.g., share price, commodity price), the higher is the price for both call and put options (and vice versa). The reason being higher implied volatility means there's greater price movement being expected by market participants, thus increased upside and downside potential.

As a general proxy, higher implied volatility (IV) indicates uncertainty in the markets that can cause swings in either direction, but it's interpreted as a bearish signal because many equity investors exit the market when there's greater uncertainty.

How is the VIX traded?

The VIX is an index and not directly tradable, similar to the S&P 500. Market participants can trade VIX futures and take a view of where the calculated index will be in the future.

Like other future products, VIX futures can be packaged into Exchange Traded Notes (ETNs) and bought by retail investors.

If an investor wants to go long on the VIX, what factors would need to be considered?

The return of being long on the VIX is affected by the futures' roll-yield. In normal times, the VIX curve is in a contango shape, where the level where I can lock VIX in the future is higher than the level today.

The return of a long VIX strategy, whether in futures directly or ETNs, needs to factor in the cost of rolling futures due to the contango shape.

What will happen to the market if North Korea invades South Korea?

This is a test of your understanding of how markets are connected and interrelated. The question to ask is: "Is the move a risk-on move or a risk-off move?" Here, political instability is a risk-off move.

Sample Response

"If North Korea invades South Korea, stocks and stock futures will fall, most notably in the direct market (say the KOSPI) and across the world. Assets would pour into safe-haven assets, US Treasuries, Gold, and currencies such as the Swiss Franc. And the prices for all these assets would increase. Yields on government bonds will go down as yields and prices are inversely related.

For Corporate bonds, the overall price performance would be mixed, with the treasury yields declining offset by widening credit spreads. In terms of safe-haven currencies, the Japanese Yen, which would typically be included and has appreciated during other times of dislocation, would likely weaken in this scenario due to the geographical proximity of Japan to the Korean peninsula and interconnected economies."

Pick one asset class or part of the market you know well and dive a bit deeper. For example, you can talk about options.

Sample Response

"I would expect implied volatility in the market to increase substantially. Demand for puts would increase, particularly for out of the money puts, and we would see skew increase significantly. Skew is the difference between the implied volatility of out of the money puts and out of the money calls, and the supply/demand imbalance in this scenario would increase skew significantly.

You could also expand into specific stocks and trading strategies you would recommend:

South Korea is a large producer of cell phone components, and this disruption would affect Apple's ability to hit its sales numbers for the next quarter. I would short Apple stock and short a basket of Apple suppliers in the delta-one market using a custom basket equity swap.

General Motors has a large manufacturing operation in Korea from its purchase of Daewoo. Their global sales numbers would be particularly affected if their Korean plants are offline for an extended period. I would short GM shares to express this view."

Could you provide some examples of how swaps can hedge fixed income risks?

Swaps are financial agreements that can hedge a variety of fixed income risks, including:

- **Interest Rate Risk:** Interest rate swaps are derivatives that enable investors to convert floating rate cash flows into fixed-rate cash flows. The swap protects against the potential bond price decline caused by when bank deposit rates rise and investors want more yield to own bonds (i.e., yields rise, prices decline). An interest rate swap refers to a forward contract that usually involves the exchange of a fixed and floating rate between two parties. The contract is agreed upon by the two borrowers having different preferences for interest rates based on their specific needs (and speculations).
- **Credit Default Risk:** Credit default risk swaps are used when investors are concerned about the bond being repaid in full and demand more yield to own bonds (yields rise, prices decline)
- **Foreign Exchange Risk:** FX swaps come into play when the bond issuer pays coupons and the bond principal in a different currency – therefore, this is used to hedge against the possibility the currency's value might fall

What are the four main types of trading on Wall Street?

1. **Proprietary Trading:** In proprietary trading, traders will purchase and sell securities using the firm's capital to make profits. Prop trading doesn't involve clients, but it functions as if it were an investment bank's internal hedge fund. However, due to regulatory changes and the Volker rule, many banks have been forced to stop prop trading – instead, most banks have spun out their prop trading desks into independent hedge funds.
2. **Flow Trading:** Today, most of the traders currently working on Wall Street are flow traders. Flow trading is when the bank acts as a principal, and the client decides if they want to buy or sell. Based on their decision, the trader will set the price and take the other side. Flow traders bring in profits through high volume transactions and charging a bid-offer spread on each transaction. A bid-offer spread involves making markets in a stock, bond, or a derivative, with the trader buying at a lower price (bid price) than they're selling it (ask price).
3. **Agency Trading:** In agency trading, the trader executes orders on behalf of a particular client. Stocks, futures, and equity options are typically agency traded on an exchange (e.g., NASDAQ, NYSE, CME). The exchange is a natural market maker, and a flow trader is rarely needed to intermediate. An exception is large-sized trades, called block trades, which often happen off-exchange and use a traditional flow trader.
4. **Electronic Trading:** Electronic trading removes human touchpoints from the trading process. The employment of salespeople and traders can be expensive, and trading in margins in certain asset classes are becoming increasingly slim. Thus, these types of automated trading platforms based on advanced algorithms are more profitable for investment banks. Electronic trading develops, sells, and supports a trading platform or algorithm. Investors can trade without having to make a call or chat with a salesperson to execute the trade. Electronic trading works best for simple, high liquidity products where there's an electronic market that could be hedged. If the platform or algorithm can connect to an exchange and trade equities or futures, electronic trading makes sense. It also works outside an exchange such as FX Spot, where market participants have moved to an electronic platform, and the algorithm can trade with other banks electronically to hedge the risk.

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What category of trading would most trades fall under today?

Most trades today are agency trades, in which traders employed at an investment bank will act as an intermediary between the investor and the exchange (and collect a commission). Based on the orders of the client (the investor), the sales trader will execute the trade as directed.

Explain how the Volcker Rule changed the sales & trading division of investment banks.

The Volcker Rule is a section under the Dodd-Frank Act that prohibits investment banks from using their accounts for short-term proprietary trading of securities and derivatives, such as futures and options. In effect, the rule stated the investment banks could no longer perform proprietary trading. However, they can make markets through flow trading. This rule was imposed because banks were becoming too involved in the monetary gains from their trading divisions instead of focusing more on agency and flow trading (i.e., clients' interests come first).

What is the “Chinese Wall” in investment banks?

In the context of an investment bank, the “Chinese Wall” refers to the barrier intended to separate the M&A and Capital Markets division from the Equity Research and Sales & Trading divisions. Given the amount of non-public information that the M&A and Capital Markets division possesses on their clients (e.g., potential acquisition news, plans for capital raises, IPOs), this separation blocks the exchange of confidential information. Thus, the Chinese Wall is an information barrier protecting Material Non-Public Information (MNPI) from passing from those working on the private-side to those on the public-side.

What is a credit default swap (CDS), and why are they used?

A credit default swap (“CDS”) is a derivative instrument that enables an investor to “swap” their credit risk with another investor. Essentially, a CDS serves as an insurance policy whereby, upon the occurrence of a credit event, the credit protection buyer will be compensated by the seller.

The buyer of a CDS is said to be buying protection (i.e., benefits if the company becomes less creditworthy and cost increases, similar to buying puts). On the other side, the seller of a CDS sells this protection, speculating that the credit will improve and can thus keep the sale price for the CDS. Later, if there's a default that must be paid out to the buyer of protection (similar to selling puts on a stock), the seller loses money.

How does the CDS market differ from the options market?

The credit default swap market (“CDS”) is similar to the options market, in which investors buy or sell options in anticipation of a share price going up or down, usually to hedge an underlying position. The difference lies in how the investor is speculating that the company's perceived creditworthiness will increase or decrease, rather than share price movement.

What is an option premium, and who receives it: the buyer or seller?

The option premium is the total amount that a buyer has paid to purchase an options contract. In all instances, the seller of the option will receive the premium.

What are the three components of an option premium?

1. **Intrinsic Value:** The intrinsic value of an option is the amount of profit investors would get if they immediately exercised the option.
2. **Time Value:** The time value of an option is the amount that an investor will pay in excess of the intrinsic value. As the expiration date nears, the time decay of the option will cause the option premium to decline.
3. **Implied Volatility (IV):** The pace of the decline will depend on the implied volatility of the stock and any potential catalysts on the horizon (e.g., investor presentation, earning report release).

If an option has more time until expiration and greater implied volatility, would you expect the premium to be greater or smaller?

The option premium would be greater since there's more time for the price of the underlying asset to increase, and more volatility means a greater likelihood of price movement in either direction.

Explain the role that time decay has in the value of an option.

Time decay, which is denoted by the option Greek theta, is the measurement of the rate of decline in the value of an option from the passage of time as the expiration date nears. As the expiration date comes closer, the effect of time decay on the option's value accelerates as there's less time for the underlying asset to change in value. However, the rate of this value erosion will be slower, and the option will retain more of its value if it's currently in-the-money (ITM).

What would you expect the theta of a long-term option to be?

A long-term contract will have a theta close to zero because it doesn't lose value each passing day, and the expiration date is far away. Theta would be significantly higher for short-term options nearing expiration, especially if it's out-of-the-money. Short-term options will lose value each day, as they have more premium to lose relative to a long-term option.

What is the relationship between delta and gamma?

- **Delta:** A measure of the expected change in an option's price resulting from a \$1.00 change in the price of the underlying asset (security or index) – used by many traders to determine the likelihood of an option being in-the-money on the expiration date.
- **Gamma:** A measure of the rate of change in the delta of an option per \$1.00 change in the price of the underlying asset – gamma is a proxy for the stability of delta.

What is the difference between American options and European options?

The major distinction is that American-style options can be exercised before the expiration date, whereas European-style options can only be exercised on the date of expiration.

For this reason, the ability to exercise an option contract before expiration makes American options more valuable than European options, all else being equal. However, whether an early exercise would have been the most profitable choice would depend on the time premium remaining on the option and the potential for events in the future that can affect its value (which cannot be predicted).

What are eurobonds?

A eurobond is a type of bond that's denominated in a currency different from the currency of the country in which the bond is being issued and sold. When spelled with a non-capitalized "e," the term "euro" refers to external rather than Europe.

What is the bid-ask spread?

The bid-ask spread is the difference between the highest price that a buyer will pay for a particular asset and the lowest price that a seller will sell it. The more illiquid an asset is, the greater the bid-ask spread will be.

What is a black swan event?

A black swan event is an unexpected, very improbable event that statistically rarely occurs, but when it does – it brings havoc and severe consequences. These events cannot be predicted and often cause catastrophic damage to the global financial markets and economy.